In [4]:

```
from IPython.display import Image, display
display(Image(filename="netflix_logo.png"))
```

Netflix Data Science Project

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

In [ ]:

```

```

1. Environment Setup and Data Loading

In this section, we import the necessary libraries for data manipulation (`pandas` , `numpy`) and visualization (`seaborn` , `matplotlib`). We also load the raw Netflix dataset.

In [5]:

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import numpy as np
```

```
from warnings import filterwarnings
filterwarnings("ignore")



df = pd.read_csv("dataset/Netflix.csv")
```

2. Initial Data Inspection

Before cleaning, we take a quick look at the first few rows of the data, its dimensions, and the data types of each column to identify potential issues.

In [6]:

```
df.head()
```

Out[6]:

	show_id	type	title	director	cast	country	date_added	release_year	rating	duration	genres
0	s1	TV Show	3%	NaN	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	14-Aug-20	2020	TV-MA	4	International TV Shows, TV Dramas, TV Sci-Fi &...
1	s10	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	15-Dec-17	2008	TV-MA	143	Horror Movies, International Movies, Thrillers
2	s100	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	5-Jan-19	2016	TV-PG	124	Dramas, International Movies, Sports Movies
3	s1000	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	1-Mar-16	2016	R	90	Comedies
4	s1001	TV Show	Blue Planet II	NaN	David Attenborough	United Kingdom	3-Dec-18	2017	TV-G	1	British TV Shows, Docuseries, Science & Nature TV

In [7]:

```
df.shape
```

Out[7]:

```
(7787, 12)
```

In [8]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7787 entries, 0 to 7786
Data columns (total 12 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
```

```
 0   show_id       7787 non-null   object
 1   type          7787 non-null   object
 2   title         7787 non-null   object
 3   director      5398 non-null   object
 4   cast          7069 non-null   object
 5   country       7280 non-null   object
 6   date_added    7777 non-null   object
 7   release_year  7787 non-null   int64
 8   rating        7780 non-null   object
 9   duration      7787 non-null   int64
 10  genres        7787 non-null   object
 11  description   7787 non-null   object
dtypes: int64(2), object(10)
memory usage: 730.2+ KB
```

In [9]:

```
df.describe()
```

Out[9]:

	release_year	duration
count	7787.000000	7787.000000
mean	2013.932580	69.122769
std	8.757395	50.950743
min	1925.000000	1.000000
25%	2013.000000	2.000000
50%	2017.000000	88.000000
75%	2018.000000	106.000000
max	2021.000000	312.000000

In [10]:

```
df.isnull().sum()
```

Out[10]:

```
show_id             0
type                0
title               0
director         2389
cast              718
country           507
date_added         10
release_year        0
rating              7
duration            0
genres              0
description         0
dtype: int64
```

In [11]:

```
df.duplicated().sum()
```

Out[11]:

```
np.int64(0)
```

3. Data Cleaning

To ensure the quality of our analysis, we perform the following:

- **Drop Unnecessary Columns:** Removing `show_id` as it serves only as a unique identifier.
- **Impute Missing Values:** Filling gaps in `director`, `cast`, and `country` with "Unknown", and using the

mode for `rating` and `date_added`.

In [12]:

```
df.drop(columns=['show_id'], inplace=True)
```

In [13]:

```
df['director'].fillna('Unknown', inplace=True)
df['cast'].fillna('Unknown', inplace=True)
df['country'].fillna('Unknown', inplace=True)
df['rating'].fillna(df['rating'].mode()[0], inplace=True)
df['date_added'].fillna(df['date_added'].mode()[0], inplace=True)
```

In [ ]:

```

```

4. Feature Engineering

We create new features to extract more value from the existing data:

1. **Date Extraction:** Converting `date_added` to datetime objects to extract year and month.
2. **Complexity Metrics:** Calculating the number of cast members and genres per title.
3. **Categorization:** Simplifying the content ratings into broader categories: **Adult**, **Teen**, and **Kids**.
4. **Localization:** Extracting the primary country of origin from the `country` column.

In [14]:

```
df['date_added'] = pd.to_datetime(df['date_added'])

df['year_added'] = df['date_added'].dt.year
df['month_added'] = df['date_added'].dt.month
```

In [15]:

```
df.drop(columns=['date_added'], inplace=True)
```

In [16]:

```
df['cast_count'] = df['cast'].apply(lambda x: len(x.split(',')))
df['genre_count'] = df['genres'].apply(lambda x: len(x.split(',')))
```

In [17]:

```
df['duration_type'] = df['type']

df.rename(columns={'duration':'duration_value'}, inplace=True)
```

In [18]:

```
df['country_main'] = df['country'].apply(lambda x: x.split(',')[0])
```

In [19]:

```
def simplify_rating(r):
    if r in ['TV-MA','R','NC-17']:
        return "Adult"
    elif r in ['PG-13','TV-14']:
        return "Teen"
    else:
        return "Kids"

df['rating_category'] = df['rating'].apply(simplify_rating)
```

In [20]:

```
df.drop(columns=['rating'], inplace=True)
```

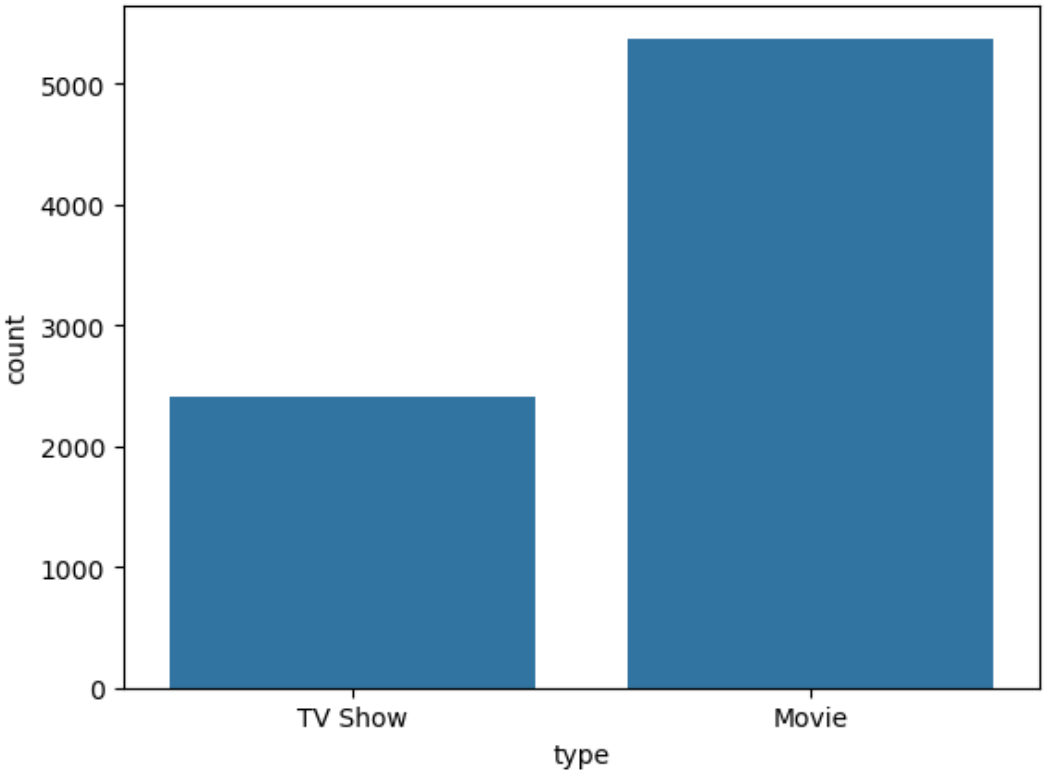
5. Exploratory Data Analysis & Visualization

With the data cleaned and enriched, we now move to the visualization phase. This help us identify:

- **The distribution of content types (Movies vs. TV Shows).**
- **Trends in content releases over the years.**
- **Relationship between content duration and its category.**

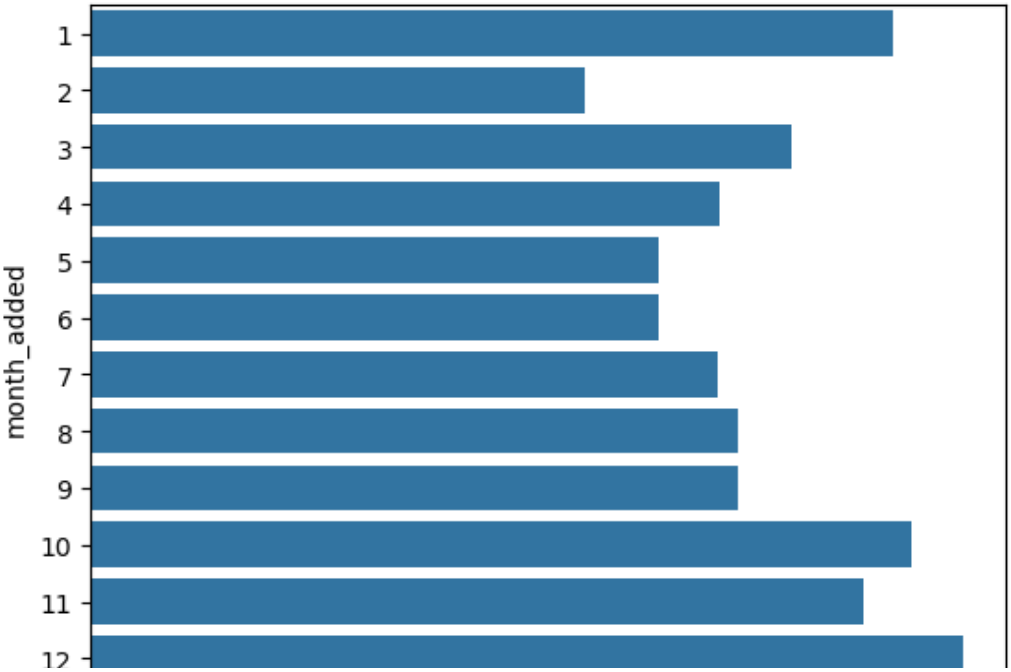
In [21]:

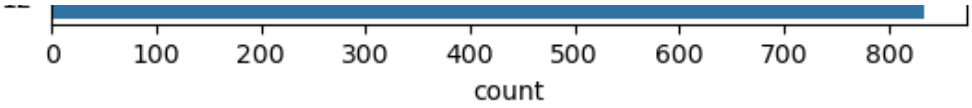
```
sns.countplot(x="type",data=df)
plt.show()
```



In [22]:

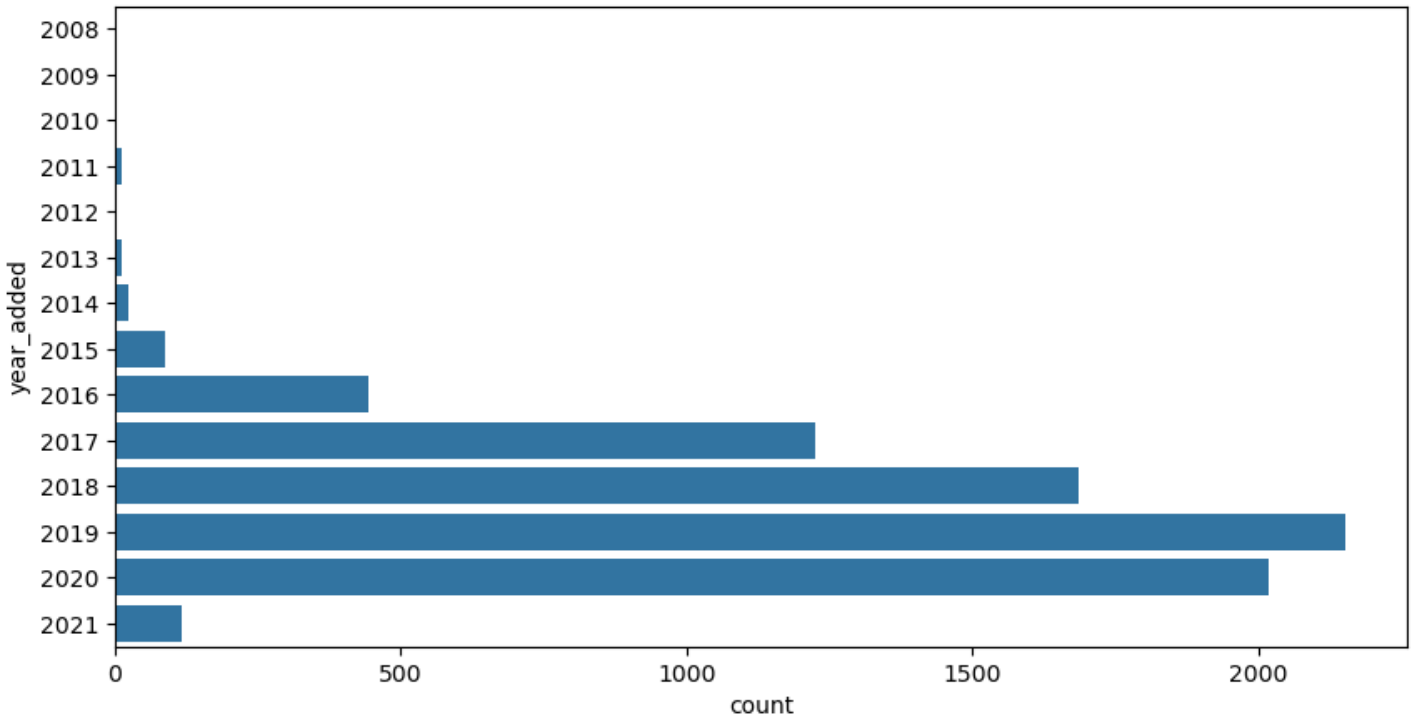
```
sns.countplot(y="month_added",data=df)
plt.show()
```





In [23]:

```
plt.figure(figsize=(10,5))
sns.countplot(y=df["year_added"])
plt.show()
```



In [ ]:

In [ ]:

In [24]:

```
df['has_director'] = df['director'].apply(lambda x: 0 if x=='Unknown' else 1)
```

In [25]:

```
df.corr(numeric_only=True)
```

Out[25]:

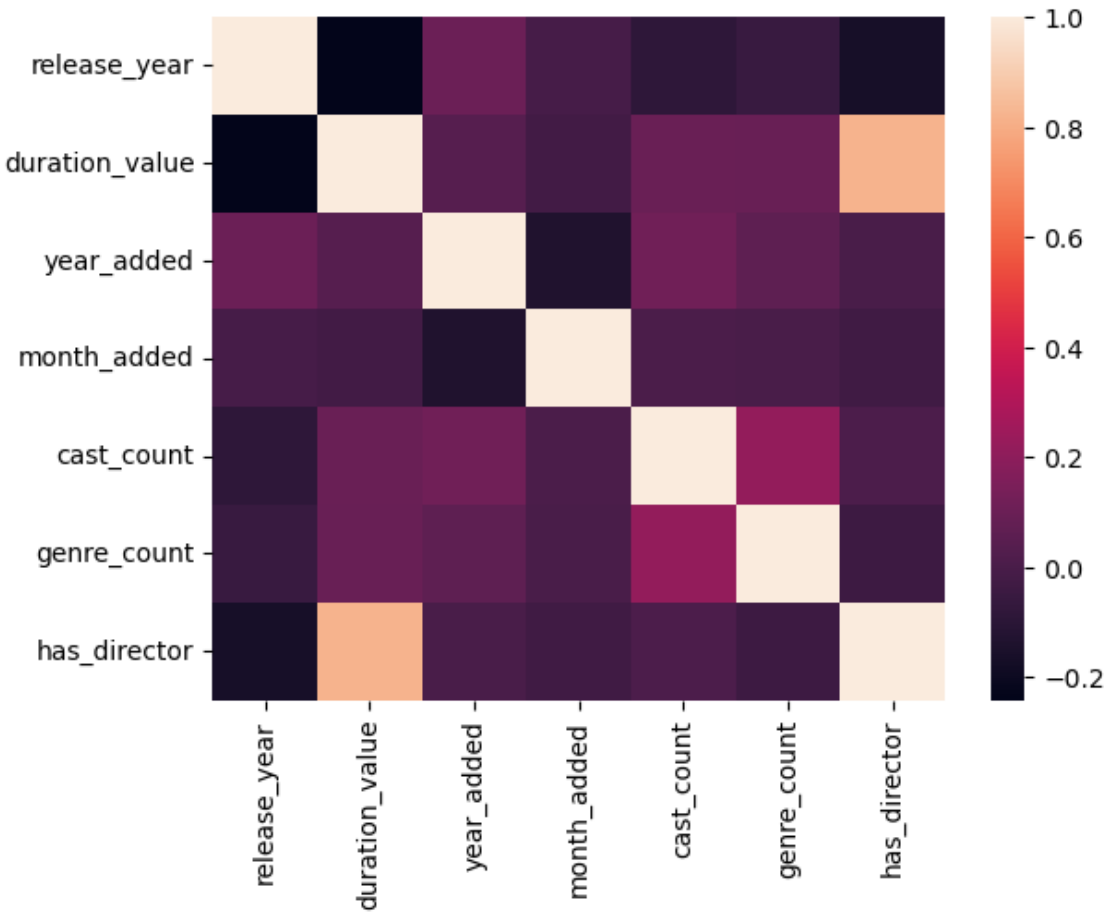
	release_year	duration_value	year_added	month_added	cast_count	genre_count	has_director
release_year	1.000000	-0.240836	0.100115	-0.009661	-0.086585	-0.052931	-0.166603
duration_value	-0.240836	1.000000	0.037102	-0.024913	0.096601	0.092256	0.820335
year_added	0.100115	0.037102	1.000000	-0.135006	0.113470	0.060552	-0.001745
month_added	-0.009661	-0.024913	-0.135006	1.000000	0.001902	-0.002332	-0.028360
cast_count	-0.086585	0.096601	0.113470	0.001902	1.000000	0.219182	0.008707
genre_count	-0.052931	0.092256	0.060552	-0.002332	0.219182	1.000000	-0.041000
has_director	-0.166603	0.820335	-0.001745	-0.028360	0.008707	-0.041000	1.000000

In [26]:

```
sns.heatmap(df.corr(numeric_only=True))
```

Out[26]:

<Axes: >



Q1] Develop a Content-Type labeling system:

In [ ]:

In [ ]:

In [ ]:

LOGISTIC - REGRESSION:

In [27]:

```
# Import necessary libraries
from sklearn.model_selection import train_test_split
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.pipeline import Pipeline
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
import pickle


num_features = ['cast_count', 'genre_count', 'duration_value']

cat_features = ['country_main', 'rating_category']

X = df[num_features + cat_features]
```

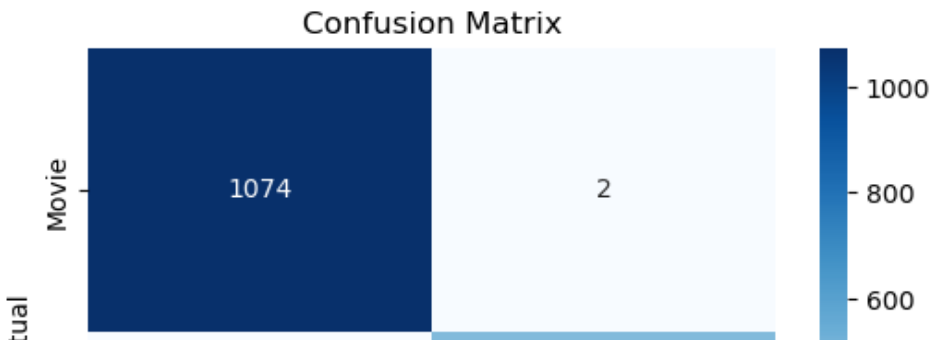
```python
y = df['type']

# Split the data into train and test sets (80-20 split)
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42
, stratify=y)


num_transformer = Pipeline(steps=[
    ('scaler', StandardScaler())
])


cat_transformer = Pipeline(steps=[
    ('onehot', OneHotEncoder(drop='first', handle_unknown='ignore'))
])


preprocessor = ColumnTransformer(
    transformers=[
        ('num', num_transformer, num_features),
        ('cat', cat_transformer, cat_features)
    ])



pipeline = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', LogisticRegression(random_state=42, max_iter=1000))
])

from sklearn.model_selection import train_test_split, GridSearchCV
param_grid = {
    'classifier__C': [0.01, 0.1, 1, 10],
    'classifier__penalty': ['l2'],
    'classifier__solver': ['lbfgs']
}

grid = GridSearchCV(
    pipeline,
    param_grid,
    cv=5,
    scoring='accuracy',
    n_jobs=-1,
    verbose=1
)

grid.fit(X_train, y_train)

best_model = grid.best_estimator_
y_pred = best_model.predict(X_test)

# Evaluate the model
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy:.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))
print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save the pipeline model (similar to laptop project)
with open('netflix_type_predictor_logisticRegression_pipeline.pkl', 'wb') as f:
    pickle.dump(best_model, f)


# Example prediction function (like app.py in laptop project)
def predict_content_type(cast_count, genre_count, duration_value,
                         country_main, rating_category):
    # Create input DataFrame
    query_df = pd.DataFrame({
        'cast_count': [cast_count],
        'genre_count': [genre_count],
        'duration_value': [duration_value],
```

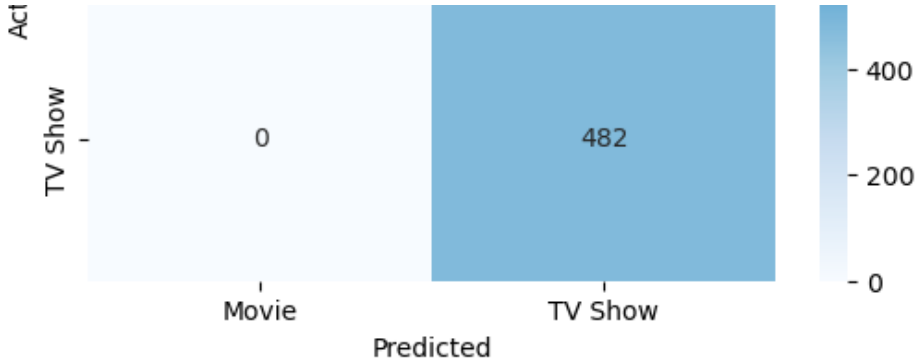
```
        'country_main': [country_main],
        'rating_category': [rating_category]
    })

    # Load the pipeline
    with open('netflix_type_predictor_logisticRegression_pipeline.pkl', 'rb') as f:
        pipe = pickle.load(f)

    # Predict
    prediction = pipe.predict(query_df)[0]
    content_type = 'TV Show' if prediction == 1 else 'Movie'
    probability = pipe.predict_proba(query_df)[0]
    conf_movie = probability[0] * 100
    conf_tv = probability[1] * 100

    return f"Predicted Type: {content_type} (Confidence: {max(conf_movie, conf_tv):.2f}%)
"

# Example usage
print("\nExample Prediction:")
result = predict_content_type(cast_count=8, genre_count=3, duration_value=143,
    country_main='India', rating_category='Adult'
)
print(result)
```

```
Fitting 5 folds for each of 4 candidates, totalling 20 fits
Accuracy: 0.9987

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      1.00      1.00      1076
     TV Show       1.00      1.00      1.00       482

    accuracy                           1.00      1558
   macro avg       1.00      1.00      1.00      1558
weighted avg       1.00      1.00      1.00      1558


Confusion Matrix:
[[1074    2]
 [   0  482]]

Example Prediction:
Predicted Type: Movie (Confidence: 100.00%)
```

In [28]:

```
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```





NAIVE - BAYES (Bernoulli) :

In [29]:

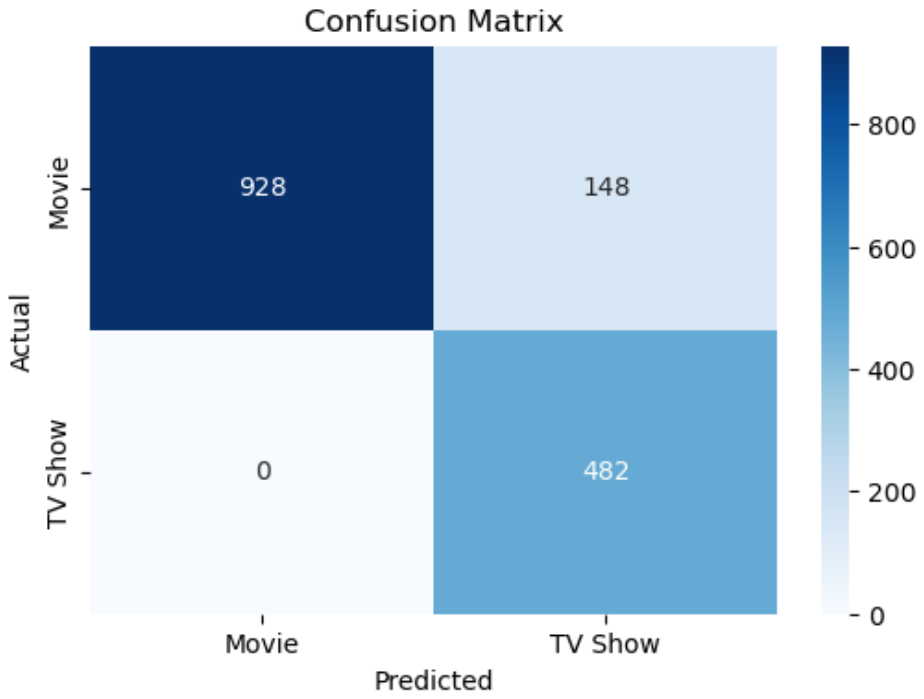
```
from sklearn.naive_bayes import BernoulliNB
from sklearn.model_selection import GridSearchCV

# Pipeline
pipeline_nb = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', BernoulliNB())
])

# □ Hyperparameter tuning for Naive Bayes
param_grid_nb = {
    'classifier__alpha': [0.1, 0.5, 1.0, 2.0],
    'classifier__binarize': [0.0, 0.5, 1.0],
    'classifier__fit_prior': [True, False]
}

grid_nb = GridSearchCV(
    pipeline_nb,
    param_grid_nb,
    cv=5,
    scoring='accuracy',
    n_jobs=-1,
    verbose=1
)

# Train
grid_nb.fit(X_train, y_train)

# Best model
best_nb_model = grid_nb.best_estimator_

print("Best Parameters (Naive Bayes):", grid_nb.best_params_)

# Predictions
y_pred = best_nb_model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
print(f"\nTuned Accuracy: {accuracy:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save tuned model
with open('netflix_type_predictor_naiveBayes_tuned.pkl', 'wb') as f:
    pickle.dump(best_nb_model, f)
```

```
Fitting 5 folds for each of 24 candidates, totalling 120 fits
Best Parameters (Naive Bayes): {'classifier__alpha': 2.0, 'classifier__binarize': 0.0, 'c
lassifier__fit_prior': False}
```

```
Tuned Accuracy: 0.9050

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.86      0.93      1076
     TV Show       0.77      1.00      0.87       482

    accuracy                           0.91      1558
   macro avg       0.88      0.93      0.90      1558
weighted avg       0.93      0.91      0.91      1558


Confusion Matrix:
[[928 148]
 [  0 482]]
```

In [30]:

```
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])

plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



KNN:

In [31]:

```
from sklearn.neighbors import KNeighborsClassifier
from sklearn.model_selection import GridSearchCV

# Pipeline
pipeline_knn = Pipeline(steps=[
    ('preprocessor', preprocessor),
    ('classifier', KNeighborsClassifier())
])
```

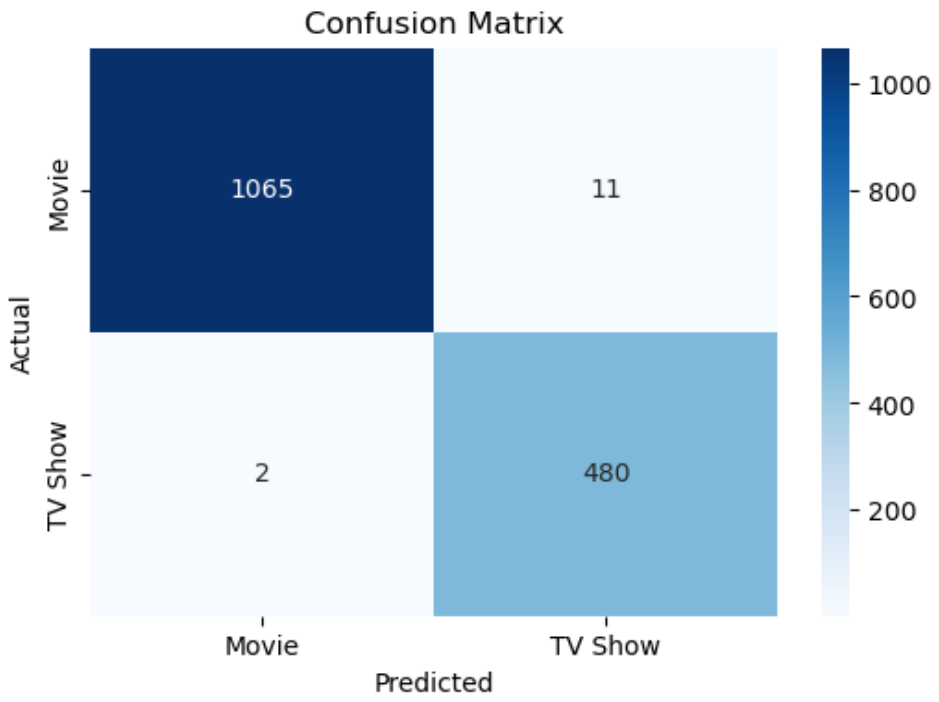
```python
# □ Hyperparameter tuning
param_grid_knn = {
    'classifier__n_neighbors': [3, 5, 7, 9, 11],
    'classifier__weights': ['uniform', 'distance'],
    'classifier__metric': ['euclidean', 'manhattan'],
    'classifier__p': [1, 2]  # 1=manhattan, 2=euclidean
}

grid_knn = GridSearchCV(
    pipeline_knn,
    param_grid_knn,
    cv=5,
    scoring='accuracy',
    n_jobs=-1,
    verbose=1
)

# Train
grid_knn.fit(X_train, y_train)

# Best model
best_knn_model = grid_knn.best_estimator_

print("Best Parameters (KNN):", grid_knn.best_params_)

# Predictions
y_pred = best_knn_model.predict(X_test)

# Evaluation
accuracy = accuracy_score(y_test, y_pred)
print(f"\nTuned Accuracy: {accuracy:.4f}")

print("\nClassification Report:")
print(classification_report(y_test, y_pred, target_names=['Movie', 'TV Show']))

print("\nConfusion Matrix:")
print(confusion_matrix(y_test, y_pred))

# Save tuned model
with open('netflix_type_predictor_knn_tuned.pkl', 'wb') as f:
    pickle.dump(best_knn_model, f)
```

```
Fitting 5 folds for each of 40 candidates, totalling 200 fits
Best Parameters (KNN): {'classifier__metric': 'euclidean', 'classifier__n_neighbors': 5,
'classifier__p': 1, 'classifier__weights': 'distance'}

Tuned Accuracy: 0.9917

Classification Report:
              precision    recall  f1-score   support

       Movie       1.00      0.99      0.99      1076
     TV Show       0.98      1.00      0.99       482

    accuracy                           0.99      1558
   macro avg       0.99      0.99      0.99      1558
weighted avg       0.99      0.99      0.99      1558


Confusion Matrix:
[[1065   11]
 [   2  480]]
```

In [32]:

```python
plt.figure(figsize=(6,4))
sns.heatmap(confusion_matrix(y_test, y_pred),
            annot=True,
            fmt='d',
            cmap='Blues',
            xticklabels=['Movie', 'TV Show'],
            yticklabels=['Movie', 'TV Show'])
```

```
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion Matrix')
plt.show()
```



In [ ]:

In [ ]:

In [ ]:

CONCLUSION: Highest accuracy is acheived through Logistic Regression.

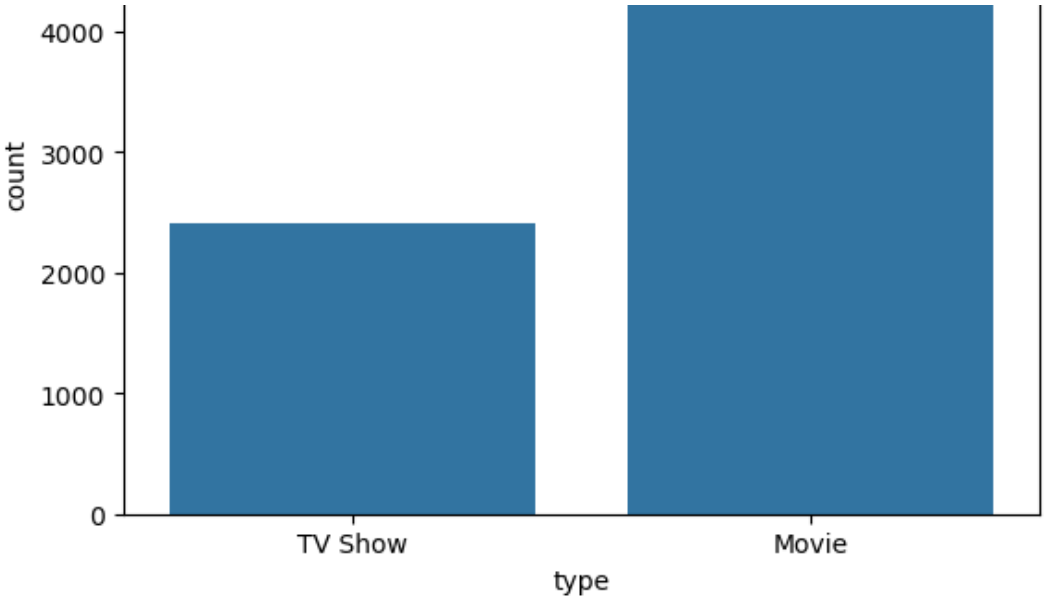
In [ ]:

In [ ]:

Q2] Predict whether a piece of content is a Movie or a TV Show based on features like cast-size, country, and release trends?

In [33]:

```
sns.countplot(x=df["type"])
plt.show()
```



In [34]:

```
data = df.groupby("type")["genre_count"].sum().reset_index()
data
```

Out[34]:

	type	genre_count
0	Movie	11546
1	TV Show	5525

In [35]:

```
data = df.groupby("type")["cast_count"].sum().reset_index()
data
```

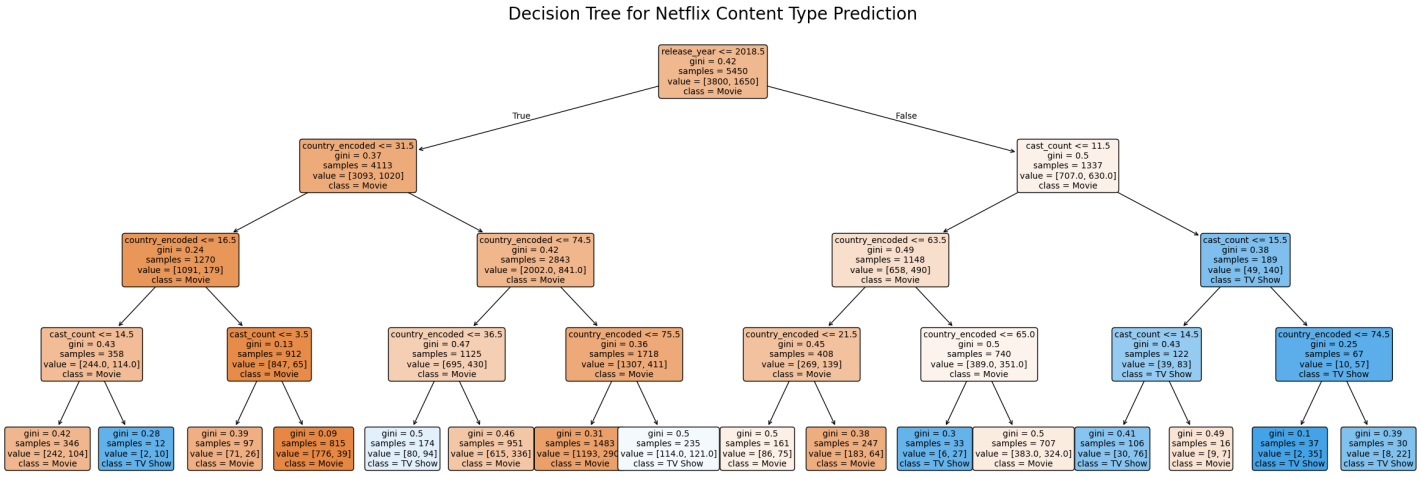
Out[35]:

	type	cast_count
0	Movie	38738
1	TV Show	17935

In [36]:

```
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier, plot_tree
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder

# 1. Preprocessing categorical features
# We extract the first country and first genre to simplify the categories
df['first_country'] = df['country'].fillna('Unknown').apply (lambda x: x.split(',')[0])

# 2. Encode strings to numbers
le_country = LabelEncoder()

df['country_encoded'] = le_country.fit_transform(df['first_country'])



feature_cols = ["cast_count", "country_encoded", "release_year"]
X = df[feature_cols]
y = df["type"]


X_train , X_test , y_train, y_test = train_test_split(X,y, test_size=0.3,random_state=42
)
```

```
model = DecisionTreeClassifier(max_depth=4, random_state=42)
model.fit(X_train , y_train)


plt.figure(figsize=(30,10))
plot_tree(model,
          feature_names=feature_cols,
          class_names=["Movie", "TV Show"],
          filled=True,
          rounded=True,
          fontsize=10,
          precision=2)

plt.title("Decision Tree for Netflix Content Type Prediction", fontsize=20)
plt.show()


print("Accuracy:",model.score(X_test, y_test))
```




```
Accuracy: 0.7064612751390672
```

Decision Tree for Netflix Content Type Prediction

The decision tree model was trained to classify Netflix content into Movies and TV Shows using features such as:

- `release_year`
- `country_encoded`
- `cast_count`

Model Insights

- **The tree splits primarily on release year, showing that newer releases tend to follow different classification patterns than older ones.**
- **Country encoding and cast count act as secondary features, refining predictions when year alone is insufficient.**
- **Leaf nodes reveal the final classification outcome, with color-coding distinguishing between predicted classes.**

Performance

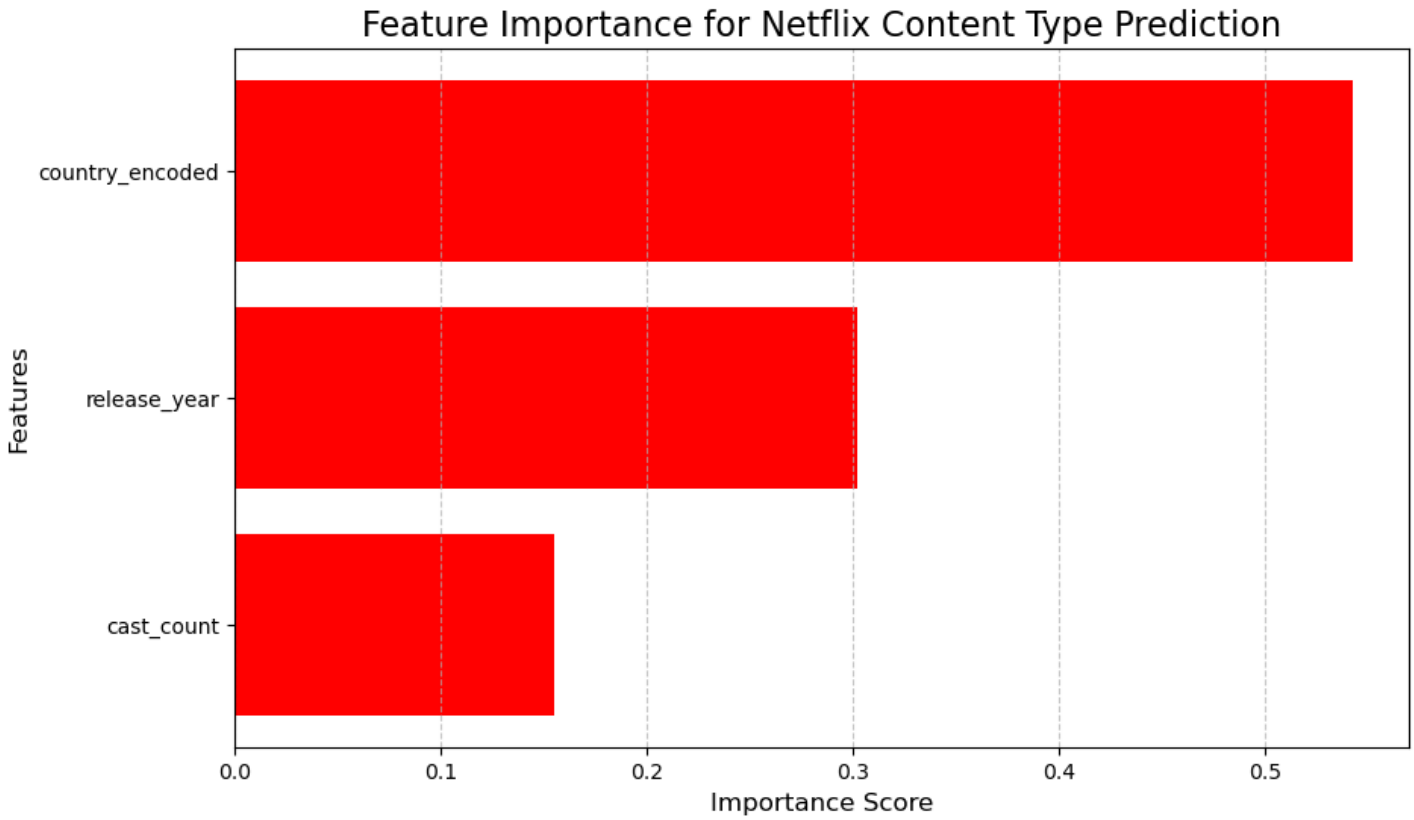
- **Accuracy: ~0.706 (≈71%)**
- **This indicates the model correctly classifies about 7 out of 10 titles.**
- **While interpretable and easy to visualize, the decision tree may suffer from overfitting and limited generalization compared to ensemble methods.**

Key Takeaway

The decision tree provides a transparent view of how Netflix content is classified, highlighting the importance of release year as the dominant feature. However, for higher accuracy and robustness, ensemble approaches (e.g., Random Forests, Gradient Boosting) would likely outperform this single-tree model.

In [37]:

```
# 1. Create a DataFrame to hold the feature names and their corresponding importance scor
es
feature_importance_df = pd.DataFrame({
    'Feature': feature_cols,
    'Importance': model.feature_importances_
}).sort_values(by='Importance', ascending=True)

# 2. Plot the Feature Importance
plt.figure(figsize=(10, 6))
plt.barh(feature_importance_df['Feature'], feature_importance_df['Importance'], color='re
d')

plt.title('Feature Importance for Netflix Content Type Prediction', fontsize=16)
plt.xlabel('Importance Score', fontsize=12)
plt.ylabel('Features', fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.7)

plt.show()

# 3. Print the values for reference
# Create the sorted dataframe and drop the old index
importance_display = feature_importance_df.sort_values(by='Importance', ascending=False)
.reset_index(drop=True)

# Display the clean table
importance_display.reset_index()
```



Out[37]:

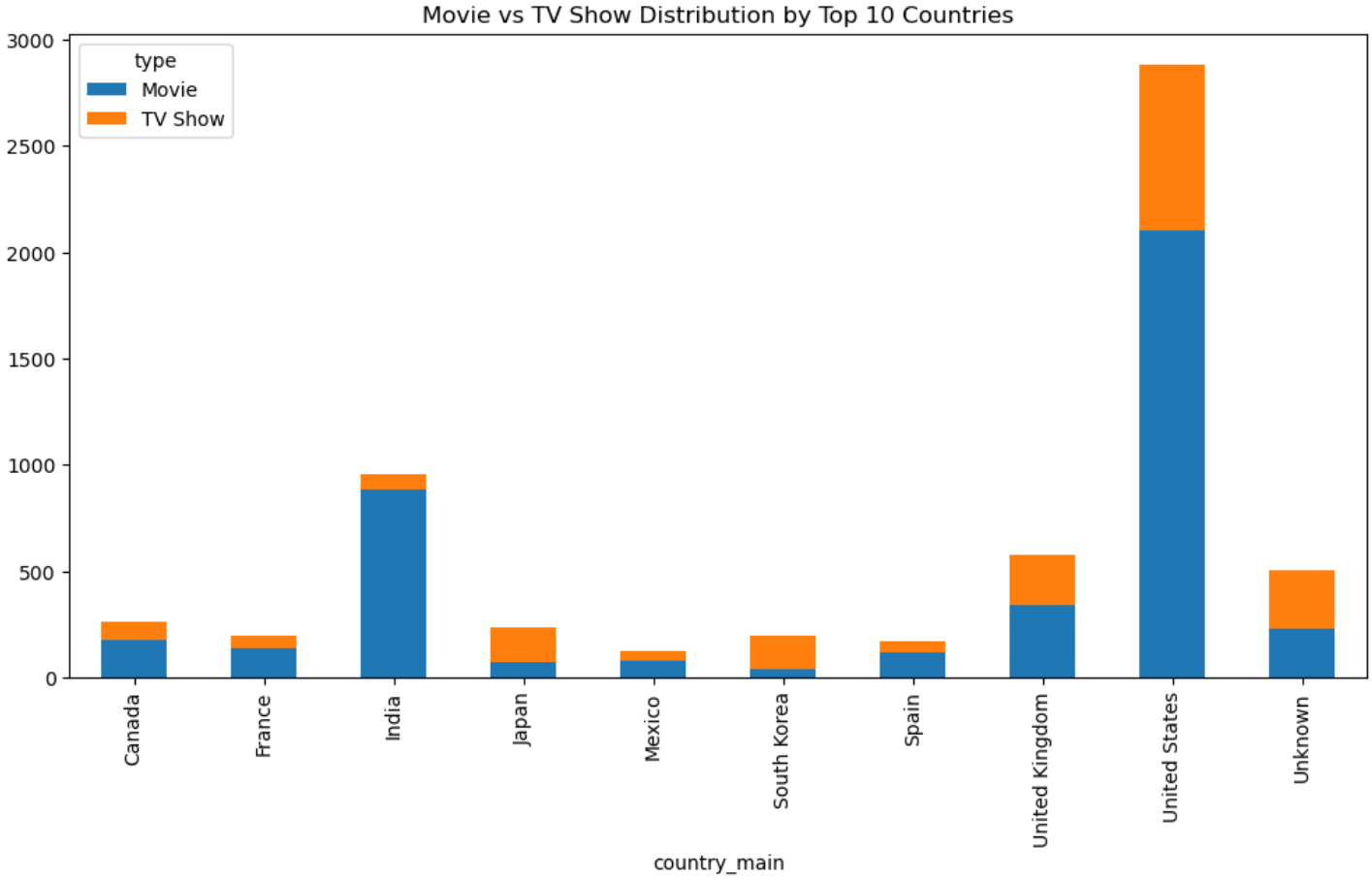
	index	Feature	Importance
0	0	country_encoded	0.542345
1	1	release_year	0.302328
2	2	cast_count	0.155326

In [ ]:

Q3] Which countries dominate Netflix content, and how does the Movie vs TV-Show ratio change across the top 10 countries?

In [38]:

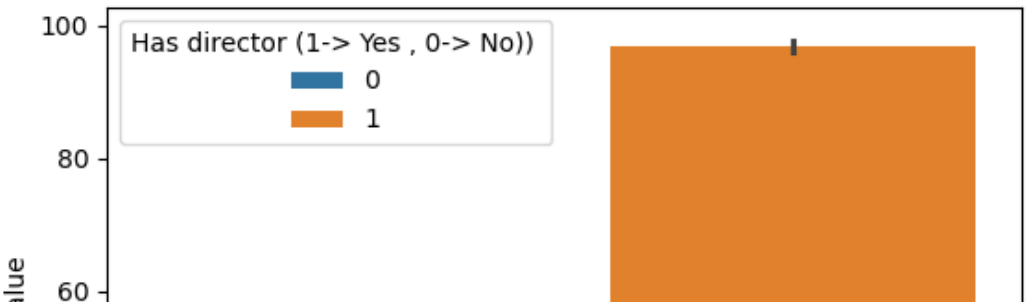
```
top10_countries = df['country_main'].value_counts().head(10).index
country_type = df[df['country_main'].isin(top10_countries)].groupby(['country_main', 'ty
pe']).size().unstack()
country_type.plot(kind='bar', stacked=True, figsize=(12,6))
plt.title('Movie vs TV Show Distribution by Top 10 Countries')
plt.show()
```

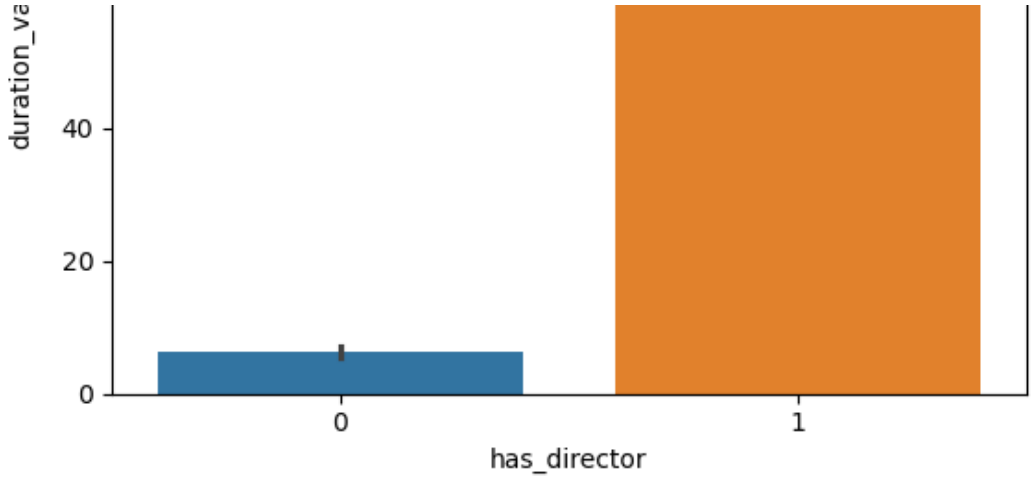


Q4] Do titles with a known director have significantly higher cast_count, genre_count, or longer duration than “Unknown” director titles?

In [39]:

```
df.groupby('has_director')[['cast_count', 'genre_count', 'duration_value']].mean().round
(2)
sns.barplot(x='has_director', y='duration_value',hue="has_director", data=df)
plt.legend(title="Has director (1-> Yes , 0-> No)) ")
plt.show()
```





In [40]:

```
df.head()
```

Out[40]:

	type	title	director	cast	country	release_year	duration_value	genres	description	year_added
0	TV Show	3%	Unknown	João Miguel, Bianca Comparato, Michel Gomes, R...	Brazil	2020	4	International TV Shows, TV Dramas, TV Sci-Fi &...	In a future where the elite inhabit an island ...	2020
1	Movie	1920	Vikram Bhatt	Rajneesh Duggal, Adah Sharma, Indraneil Sengup...	India	2008	143	Horror Movies, International Movies, Thrillers	An architect and his wife move into a castle t...	2017
2	Movie	3 Heroines	Iman Brotoseno	Reza Rahadian, Bunga Citra Lestari, Tara Basro...	Indonesia	2016	124	Dramas, International Movies, Sports Movies	Three Indonesian women break records by becomi...	2019
3	Movie	Blue Mountain State: The Rise of Thadland	Lev L. Spiro	Alan Ritchson, Darin Brooks, James Cade, Rob R...	United States	2016	90	Comedies	New NFL star Thad buys his old teammates' belo...	2016
4	TV Show	Blue Planet II	Unknown	David Attenborough	United Kingdom	2017	1	British TV Shows, Docuseries, Science & Nature TV	This sequel to the award-winning nature series...	2018

Q5] Develop a Content- recommendation system:

first we will use Word-Embedding:

In [41]:

```
import pandas as pd
import pickle
import numpy as np
from nltk.corpus import stopwords
```

```python
from nltk.tokenize import word_tokenize
from gensim.models import Word2Vec
from sklearn.metrics.pairwise import cosine_similarity

# ----------------------------
# STEP 1: Create tags (same as before)
# ----------------------------
df['tags'] = (
    df['title'].fillna('') + " " +
    df['genres'].fillna('') + " " +
    df['description'].fillna('') + " " +
    df['cast'].fillna('') + " " +
    df['director'].fillna('') + " " +
    df['country_main'].fillna('') + " " +
    df['rating_category'].fillna('') + " " +
    df['type'].fillna('')
)

df['tags'] = df['tags'].str.lower()

# ----------------------------
# STEP 2: Tokenization
# ----------------------------
stop_words = set(stopwords.words('english'))

df['tokens'] = df['tags'].apply(
    lambda x: [word for word in word_tokenize(x)
               if word.isalnum() and word not in stop_words]
)


# ----------------------------
# STEP 3: Train Word2Vec Model
# ----------------------------
w2v_model = Word2Vec(
    sentences=df['tokens'],
    vector_size=100,
    window=5,
    min_count=1,
    workers=4
)

# ----------------------------
# STEP 4: Convert each content into vector
# ----------------------------
def get_doc_vector(tokens):
    vectors = []
    for word in tokens:
        if word in w2v_model.wv:
            vectors.append(w2v_model.wv[word])

    if len(vectors) == 0:
        return np.zeros(100)

    return np.mean(vectors, axis=0)

df['vector'] = df['tokens'].apply(get_doc_vector)

# Convert to matrix
vectors = np.vstack(df['vector'].values)

# ----------------------------
# STEP 5: Recommendation Class
# ----------------------------
class RecommendationModel:
    def __init__(self, vectors, df, w2v_model):
        self.vectors = vectors
        self.df = df.reset_index(drop=True)
        self.w2v_model = w2v_model

    def text_to_vector(self, text):
        tokens = word_tokenize(text.lower())
```

```python
        return get_doc_vector(tokens)

    def recommend(self, content, top_n=10):
        mask = self.df['title'].str.contains(content, case=False, na=False)

        if mask.any():
            idx = self.df[mask].index[0]
            query_vec = self.vectors[idx].reshape(1, -1)
            print(" Found exact title → using full features")
        else:
            query_vec = self.text_to_vector(content).reshape(1, -1)

        similarity = cosine_similarity(query_vec, self.vectors)[0]

        content_list = sorted(
            list(enumerate(similarity)),
            reverse=True,
            key=lambda x: x[1]
        )[1:top_n+1]

        print(f"\n\nRecommendations similar to: **{content}**\n")

        for rank, (i, score) in enumerate(content_list, 1):
            row = self.df.iloc[i]
            print(f"{rank:2}) • {row['title']}  "
                  f"| {row['type']}  "
                  f"| {row['genres'][:70]}...")

# ----------------------------
# STEP 6: Save & Load Model
# ----------------------------
model = RecommendationModel(vectors, df, w2v_model)

with open("content_recommendation_word2vec.pkl", "wb") as f:
    pickle.dump(model, f)

with open("content_recommendation_word2vec.pkl", "rb") as f:
    loaded_model = pickle.load(f)

# ----------------------------
# STEP 7: Test
# ----------------------------
loaded_model.recommend("Kesari 2")
loaded_model.recommend("Dune")
loaded_model.recommend("Stranger Things")
loaded_model.recommend("Article 15")
loaded_model.recommend("Raone")
loaded_model.recommend("3 Idiots")
loaded_model.recommend("Dhurandhar")
loaded_model.recommend("Dhurandhar - The Revenge")
```

```
Recommendations similar to: **Kesari 2**

 1) • Mukhsin  | Movie  | Children & Family Movies, Comedies, Dramas...
 2) • The Breadwinner  | Movie  | Children & Family Movies, Dramas, International Movies.
..
 3) • Magnus  | Movie  | Children & Family Movies, Documentaries, International Movies...
 4) • The Rainbow Troops  | Movie  | Children & Family Movies, Dramas, International Movi
es...
 5) • Luccas Neto in: Children's Day  | Movie  | Children & Family Movies, Comedies, Musi
c & Musicals...
 6) • Misfit 2  | Movie  | Children & Family Movies, Comedies, International Movies...
 7) • Tee Shot: Ariya Jutanugarn  | Movie  | Children & Family Movies, Dramas, Internatio
nal Movies...
 8) • Nang Nak  | Movie  | Horror Movies, International Movies, Romantic Movies...
 9) • Garuda in My Heart 2  | Movie  | Children & Family Movies, Dramas, Sports Movies...
10) • Yes or No  | Movie  | International Movies, LGBTQ Movies, Romantic Movies...


Recommendations similar to: **Dune**
```

```
 1) • 1920  | Movie  | Horror Movies, International Movies, Thrillers...
 2) • 3 Heroines  | Movie  | Dramas, International Movies, Sports Movies...
 3) • Blue Mountain State: The Rise of Thadland  | Movie  | Comedies...
 4) • Blue Planet II  | TV Show  | British TV Shows, Docuseries, Science & Nature TV...
 5) • Blue Ruin  | Movie  | Independent Movies, Thrillers...
 6) • Blue Streak  | Movie  | Action & Adventure, Comedies...
 7) • Blue Valentine  | Movie  | Dramas, Independent Movies, Romantic Movies...
 8) • BluffMaster!  | Movie  | Comedies, International Movies, Romantic Movies...
 9) • Blurred Lines: Inside the Art World  | Movie  | Documentaries...
10) • BNA  | TV Show  | Anime Series, International TV Shows...
 Found exact title → using full features


Recommendations similar to: **Stranger Things**

 1) • Selena: The Series  | TV Show  | TV Dramas...
 2) • The OA  | TV Show  | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy...
 3) • Brews Brothers  | TV Show  | TV Comedies...
 4) • Chip and Potato  | TV Show  | Kids' TV...
 5) • Messiah  | TV Show  | TV Dramas, TV Thrillers...
 6) • Kipo and the Age of Wonderbeasts  | TV Show  | Kids' TV, TV Comedies...
 7) • LEGO Friends: The Power of Friendship  | TV Show  | Kids' TV...
 8) • Wrong Kind of Black  | TV Show  | TV Comedies, TV Dramas...
 9) • Riverdale  | TV Show  | Crime TV Shows, TV Dramas, TV Mysteries...
10) • The Haunting of Hill House  | TV Show  | TV Dramas, TV Horror, TV Mysteries...
 Found exact title → using full features


Recommendations similar to: **Article 15**

 1) • Uriyadi 2  | Movie  | Dramas, International Movies...
 2) • Thackeray (Hindi)  | Movie  | Dramas, International Movies...
 3) • Hazaaron Khwaishein Aisi  | Movie  | Dramas, International Movies, Romantic Movies.
..
 4) • Super Nani  | Movie  | Comedies, Dramas, International Movies...
 5) • Minnaminugu the FireFly  | Movie  | Dramas, International Movies...
 6) • Anthony Kaun Hai?  | Movie  | Comedies, Dramas, International Movies...
 7) • Harishchandrachi Factory  | Movie  | Dramas, International Movies...
 8) • Axone  | Movie  | Comedies, Dramas, Independent Movies...
 9) • Duplicate  | Movie  | Comedies, Dramas, International Movies...
10) • Bhool Bhulaiyaa  | Movie  | Dramas, International Movies, Thrillers...


Recommendations similar to: **Raone**

 1) • 1920  | Movie  | Horror Movies, International Movies, Thrillers...
 2) • 3 Heroines  | Movie  | Dramas, International Movies, Sports Movies...
 3) • Blue Mountain State: The Rise of Thadland  | Movie  | Comedies...
 4) • Blue Planet II  | TV Show  | British TV Shows, Docuseries, Science & Nature TV...
 5) • Blue Ruin  | Movie  | Independent Movies, Thrillers...
 6) • Blue Streak  | Movie  | Action & Adventure, Comedies...
 7) • Blue Valentine  | Movie  | Dramas, Independent Movies, Romantic Movies...
 8) • BluffMaster!  | Movie  | Comedies, International Movies, Romantic Movies...
 9) • Blurred Lines: Inside the Art World  | Movie  | Documentaries...
10) • BNA  | TV Show  | Anime Series, International TV Shows...
 Found exact title → using full features


Recommendations similar to: **3 Idiots**

 1) • Tum Milo Toh Sahi  | Movie  | Comedies, Dramas, International Movies...
 2) • Chal Bhaag  | Movie  | Comedies, Dramas, International Movies...
 3) • Welcome to Sajjanpur  | Movie  | Comedies, Dramas, International Movies...
 4) • Dedh Ishqiya  | Movie  | Comedies, Dramas, International Movies...
 5) • Panipat - The Great Betrayal  | Movie  | Dramas, International Movies, Romantic Mov
ies...
 6) • Pad Man  | Movie  | Dramas, International Movies, Romantic Movies...
 7) • Ungli  | Movie  | Comedies, Dramas, International Movies...
 8) • Hasee Toh Phasee  | Movie  | Comedies, Dramas, International Movies...
 9) • Eh Janam Tumhare Lekhe  | Movie  | Dramas, International Movies...
10) • EMI: Liya Hai To Chukana Padega  | Movie  | Comedies, Dramas, International Movies.
..
```

```
Recommendations similar to: **Dhurandhar**

 1) • 1920  | Movie  | Horror Movies, International Movies, Thrillers...
 2) • 3 Heroines  | Movie  | Dramas, International Movies, Sports Movies...
 3) • Blue Mountain State: The Rise of Thadland  | Movie  | Comedies...
 4) • Blue Planet II  | TV Show  | British TV Shows, Docuseries, Science & Nature TV...
 5) • Blue Ruin  | Movie  | Independent Movies, Thrillers...
 6) • Blue Streak  | Movie  | Action & Adventure, Comedies...
 7) • Blue Valentine  | Movie  | Dramas, Independent Movies, Romantic Movies...
 8) • BluffMaster!  | Movie  | Comedies, International Movies, Romantic Movies...
 9) • Blurred Lines: Inside the Art World  | Movie  | Documentaries...
10) • BNA  | TV Show  | Anime Series, International TV Shows...


Recommendations similar to: **Dhurandhar - The Revenge**

 1) • May the Devil Take You  | Movie  | Horror Movies, International Movies...
 2) • Camarón: The Film  | Movie  | Documentaries, International Movies, Music & Musicals
...
 3) • Bigflo & Oli: Hip Hop Frenzy  | Movie  | Documentaries, International Movies, Music
& Musicals...
 4) • Avicii: True Stories  | Movie  | Documentaries, International Movies, Music & Music
als...
 5) • 7 Din Mohabbat In  | Movie  | Comedies, International Movies, Music & Musicals...
 6) • Ajab Prem Ki Ghazab Kahani  | Movie  | Comedies, International Movies, Music & Musi
cals...
 7) • Kahaani  | Movie  | International Movies, Music & Musicals, Thrillers...
 8) • The Breakup Playlist  | Movie  | Dramas, International Movies, Music & Musicals...
 9) • Kismat Konnection  | Movie  | Comedies, International Movies, Music & Musicals...
10) • Yoko and His Friends  | Movie  | Children & Family Movies...
```

now we will use TF-IDF:

In [42]:

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import pickle


tfidf = TfidfVectorizer(
    stop_words='english',
    max_features=5000,      # limit to most important words
    ngram_range=(1, 2)      # also consider two-word phrases like "sci-fi"
)

tfidf_matrix = tfidf.fit_transform(df['tags'])

# 1. Simple Recommendation Class (much cleaner than Word2Vec)
class RecommendationModel:
    def __init__(self, tfidf_matrix, df, tfidf_vectorizer):
        self.tfidf_matrix = tfidf_matrix
        self.df = df.reset_index(drop=True)
        self.tfidf = tfidf_vectorizer

    def recommend(self, content, top_n=10):
        # Try exact title match first
        mask = self.df['title'].str.contains(content, case=False, na=False)
        if mask.any():
            idx = self.df[mask].index[0]
            query_vec = self.tfidf_matrix[idx]
            print("Found exact title - using full features")
        else:
            # Convert user search to TF-IDF vector
            query_vec = self.tfidf.transform([content.lower()])

        # Compute similarity
```

```
        similarity = cosine_similarity(query_vec, self.tfidf_matrix).flatten()

        # Get top N (excluding itself)
        similar_indices = similarity.argsort()[::-1][1:top_n+1]

        print(f"\n\nRecommendations similar to: **{content}**\n")
        for rank, i in enumerate(similar_indices, 1):
            row = self.df.iloc[i]
            print(f"{rank:2} • {row['title']} | {row['type']} | {row['genres'][:80]}..."
)

# 2. Train and save
model = RecommendationModel(tfidf_matrix, df, tfidf)
with open("content_recommendation_tfidf.pkl", "wb") as f:
    pickle.dump(model, f)

# 3. Load and test:
with open("content_recommendation_tfidf.pkl", "rb") as f:
    loaded_model = pickle.load(f)

loaded_model.recommend("Kesari 2")
loaded_model.recommend("Dune")
loaded_model.recommend("Stranger Things")
loaded_model.recommend("Article 15")
loaded_model.recommend("Raone")
loaded_model.recommend("3 Idiots")
loaded_model.recommend("Dhurandhar")
loaded_model.recommend("Dhurandhar - The Revenge")
```

```
Recommendations similar to: **Kesari 2**

 1 • Blue Jay | Movie | Dramas, Independent Movies, Romantic Movies...
 2 • Blue Jasmine | Movie | Comedies, Dramas, Independent Movies...
 3 • Blue Is the Warmest Color | Movie | Dramas, Independent Movies, International Movies
...
 4 • Blue Exorcist | TV Show | Anime Series, International TV Shows...
 5 • Blown Away | TV Show | International TV Shows, Reality TV...
 6 • Blow | Movie | Dramas...
 7 • Bloodride | TV Show | International TV Shows, TV Horror, TV Mysteries...
 8 • Bloodline | TV Show | TV Dramas, TV Mysteries, TV Thrillers...
 9 • Blood Will Tell | Movie | Dramas, Independent Movies, International Movies...
10 • Blood Pact | TV Show | Crime TV Shows, International TV Shows, TV Dramas...


Recommendations similar to: **Dune**

 1 • Blue Jay | Movie | Dramas, Independent Movies, Romantic Movies...
 2 • Blue Jasmine | Movie | Comedies, Dramas, Independent Movies...
 3 • Blue Is the Warmest Color | Movie | Dramas, Independent Movies, International Movies
...
 4 • Blue Exorcist | TV Show | Anime Series, International TV Shows...
 5 • Blown Away | TV Show | International TV Shows, Reality TV...
 6 • Blow | Movie | Dramas...
 7 • Bloodride | TV Show | International TV Shows, TV Horror, TV Mysteries...
 8 • Bloodline | TV Show | TV Dramas, TV Mysteries, TV Thrillers...
 9 • Blood Will Tell | Movie | Dramas, Independent Movies, International Movies...
10 • Blood Pact | TV Show | Crime TV Shows, International TV Shows, TV Dramas...
Found exact title - using full features


Recommendations similar to: **Stranger Things**

 1 • Beyond Stranger Things | TV Show | Stand-Up Comedy & Talk Shows, TV Mysteries, TV Sc
i-Fi & Fantasy...
 2 • Nightflyers | TV Show | TV Horror, TV Mysteries, TV Sci-Fi & Fantasy...
 3 • Helix | TV Show | TV Horror, TV Mysteries, TV Sci-Fi & Fantasy...
 4 • Warrior Nun | TV Show | TV Action & Adventure, TV Mysteries, TV Sci-Fi & Fantasy...
 5 • The Umbrella Academy | TV Show | TV Action & Adventure, TV Mysteries, TV Sci-Fi & Fa
ntasy...
 6 • The 4400 | TV Show | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy...
 7 • Chilling Adventures of Sabrina | TV Show | TV Horror, TV Mysteries, TV Sci-Fi & Fant
```

asy...
8 • The Vampire Diaries | TV Show | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy...
9 • Anjaan: Special Crimes Unit | TV Show | International TV Shows, TV Horror, TV Mysteries...
10 • The Messengers | TV Show | TV Dramas, TV Mysteries, TV Sci-Fi & Fantasy...
Found exact title - using full features

Recommendations similar to: **Article 15**

1 • Vanjagar Ulagam | Movie | Dramas, International Movies, Thrillers...
2 • Ittefaq | Movie | Dramas, International Movies, Thrillers...
3 • Oththa Seruppu Size 7 | Movie | Dramas, International Movies, Thrillers...
4 • Ginny Weds Sunny | Movie | Comedies, International Movies, Romantic Movies...
5 • Traffic Signal | Movie | Dramas, Independent Movies, International Movies...
6 • Issaq | Movie | Dramas, International Movies, Music & Musicals...
7 • Pink | Movie | Dramas, International Movies...
8 • Jonaki | Movie | Dramas, Independent Movies, International Movies...
9 • W/O Ram | Movie | Dramas, International Movies, Thrillers...
10 • One Day: Justice Delivered | Movie | Dramas, International Movies...

Recommendations similar to: **Raone**

1 • Blue Jay | Movie | Dramas, Independent Movies, Romantic Movies...
2 • Blue Jasmine | Movie | Comedies, Dramas, Independent Movies...
3 • Blue Is the Warmest Color | Movie | Dramas, Independent Movies, International Movies...
4 • Blue Exorcist | TV Show | Anime Series, International TV Shows...
5 • Blown Away | TV Show | International TV Shows, Reality TV...
6 • Blow | Movie | Dramas...
7 • Bloodride | TV Show | International TV Shows, TV Horror, TV Mysteries...
8 • Bloodline | TV Show | TV Dramas, TV Mysteries, TV Thrillers...
9 • Blood Will Tell | Movie | Dramas, Independent Movies, International Movies...
10 • Blood Pact | TV Show | Crime TV Shows, International TV Shows, TV Dramas...
Found exact title - using full features

Recommendations similar to: **3 Idiots**

1 • PK | Movie | Comedies, Dramas, International Movies...
2 • Ek Main Aur Ekk Tu | Movie | Comedies, International Movies, Romantic Movies...
3 • Talaash | Movie | Dramas, International Movies, Thrillers...
4 • Taare Zameen Par | Movie | Dramas, International Movies...
5 • Dil | Movie | Comedies, Dramas, International Movies...
6 • Gori Tere Pyaar Mein | Movie | Comedies, Dramas, International Movies...
7 • The Legend of Michael Mishra | Movie | Comedies, Dramas, International Movies...
8 • Unbroken | Movie | Documentaries, International Movies...
9 • Andaz Apna Apna | Movie | Action & Adventure, Comedies, Cult Movies...
10 • Don | Movie | Action & Adventure, International Movies...

Recommendations similar to: **Dhurandhar**

1 • Blue Jay | Movie | Dramas, Independent Movies, Romantic Movies...
2 • Blue Jasmine | Movie | Comedies, Dramas, Independent Movies...
3 • Blue Is the Warmest Color | Movie | Dramas, Independent Movies, International Movies...
4 • Blue Exorcist | TV Show | Anime Series, International TV Shows...
5 • Blown Away | TV Show | International TV Shows, Reality TV...
6 • Blow | Movie | Dramas...
7 • Bloodride | TV Show | International TV Shows, TV Horror, TV Mysteries...
8 • Bloodline | TV Show | TV Dramas, TV Mysteries, TV Thrillers...
9 • Blood Will Tell | Movie | Dramas, Independent Movies, International Movies...
10 • Blood Pact | TV Show | Crime TV Shows, International TV Shows, TV Dramas...

Recommendations similar to: **Dhurandhar - The Revenge**

1 • The Car: Road to Revenge | Movie | Action & Adventure, Horror Movies, Sci-Fi & Fantasy...
2 • Naan Sigappu Manithan | Movie | Action & Adventure, Dramas, International Movies...

```
 3 • Uma Maheswara Ugra Roopasya | Movie | Comedies, Dramas, Independent Movies...
 4 • Roxanne Roxanne | Movie | Dramas...
 5 • In Line | Movie | Dramas, International Movies...
 6 • Shuddhi | Movie | Dramas, International Movies, Thrillers...
 7 • Manu | Movie | Dramas, Independent Movies, International Movies...
 8 • Lady J | Movie | Dramas, Independent Movies, International Movies...
 9 • Mr. Chandramouli | Movie | Dramas, International Movies, Thrillers...
10 • Kaalia | Movie | Action & Adventure, Dramas, International Movies...
```

Q6] Can we forecast how many titles Netflix will add in 2022–2025?

In [43]:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

from sklearn.model_selection import train_test_split
from sklearn.metrics import r2_score, mean_absolute_error

yearly_data = df.groupby(['year_added','genre_count']).size().reset_index(name='title_co
unt')
yearly_data = yearly_data[yearly_data['year_added'] <= 2021]

X = yearly_data[['year_added','genre_count']]
y = yearly_data['title_count']


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

future_years = pd.DataFrame({
    'year_added': [2022, 2023, 2024, 2025],
    'genre_count': [yearly_data['genre_count'].mean()] * 4
})
```

a) Linear Regression:

In [44]:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler, PolynomialFeatures
from sklearn.linear_model import LinearRegression

linear_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('poly', PolynomialFeatures(degree=1)),
    ('model', LinearRegression())
])

linear_pipeline.fit(X_train, y_train)

y_pred_linear = linear_pipeline.predict(X_test)
linear_forecast = linear_pipeline.predict(future_years)
```

b) Adaboost:

In [45]:

```
from sklearn.ensemble import AdaBoostRegressor
from sklearn.tree import DecisionTreeRegressor

ada_pipeline = Pipeline([
    ('model', AdaBoostRegressor(
        estimator=DecisionTreeRegressor(max_depth=6),
        n_estimators=200,
```

```
        learning_rate=0.5,
        random_state=42
    ))
])

ada_pipeline.fit(X_train, y_train)

y_pred_ada = ada_pipeline.predict(X_test)
ada_forecast = ada_pipeline.predict(future_years)
```

c) Decision tree:

In [46]:

```
from sklearn.tree import DecisionTreeRegressor

tree_pipeline = Pipeline([
    ('model', DecisionTreeRegressor(
        max_depth=3,
        random_state=42
    ))
])

tree_pipeline.fit(X_train, y_train)

y_pred_tree = tree_pipeline.predict(X_test)
tree_forecast = tree_pipeline.predict(future_years)
```

d) Extratrees:

In [47]:

```
from sklearn.ensemble import ExtraTreesRegressor
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error


et_pipeline = Pipeline([
    ('model', ExtraTreesRegressor(
        n_estimators=200,
        max_depth=5,
        random_state=42,
        n_jobs=-1
    ))
])

et_pipeline.fit(X_train, y_train)

y_pred_et = et_pipeline.predict(X_test)
et_forecast = et_pipeline.predict(future_years)
```

e) Support Vector Regression (SVR):

In [48]:

```
from sklearn.svm import SVR
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error

svr_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('model', SVR(kernel='rbf'))
])

svr_pipeline.fit(X_train, y_train)
y_pred_svr = svr_pipeline.predict(X_test)
svr_forecast = svr_pipeline.predict(future_years)
```

f) Gradient Boosting Regression:

In [49]:

```
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.pipeline import Pipeline
from sklearn.metrics import r2_score, mean_absolute_error


gb_pipeline = Pipeline([
    ('model', GradientBoostingRegressor(
        n_estimators=100,
        learning_rate=0.1,
        max_depth=3,
        random_state=42
    ))
])

gb_pipeline.fit(X_train, y_train)

y_pred_gb = gb_pipeline.predict(X_test)
gb_forecast = gb_pipeline.predict(future_years)
```

g) Polynomial Regression:

In [50]:

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import PolynomialFeatures
from sklearn.linear_model import LinearRegression

poly_pipeline = Pipeline([
    ('poly', PolynomialFeatures(degree=2)),
    ('model', LinearRegression())
])

poly_pipeline.fit(X_train, y_train)

y_pred_poly = poly_pipeline.predict(X_test)
poly_forecast = poly_pipeline.predict(future_years)
```

In [ ]:

```

```

In [51]:

```
forecast_df = pd.DataFrame({
    'Year': [2022, 2023, 2024, 2025],
    'Linear': linear_forecast.astype(int),
    'Polynomial Regression':poly_forecast.astype(int),
    'AdaBoost': ada_forecast.astype(int),
    'Decision Tree': tree_forecast.astype(int),
    'Extra Trees': et_forecast.astype(int),
    'Support-Vector Regression':svr_forecast.astype(int),
    'Gradient-Boosting Regression':gb_forecast.astype(int)
})

forecast_df.reset_index()
```

Out[51]:

	index	Year	Linear	Polynomial Regression	AdaBoost	Decision Tree	Extra Trees	Support-Vector Regression	Gradient-Boosting Regression
0	0	2022	536	639	39	28	177	28	39
1	1	2023	583	721	39	28	177	27	39

2	2	2024	630	808	39	28	177	26	39
	index	Year	Linear	Polynomial Regression	AdaBoost	Decision Tree	Extra Trees	Support-Vector Regression	Gradient-Boosting Regression
3	3	2025	677	899	39	28	177	25	39

In [52]:

```
print("\n\n\n")


tolerance=20

results = pd.DataFrame({
    'Model': [
        'Linear',
        'Polynomial Regression',
        'AdaBoost',
        'Decision Tree',
        'Extra Trees',
        'Support-Vector Regression',
        'Gradient-Boosting Regression'
    ],
    'R2 Score': [
        r2_score(y_test, y_pred_linear),
        r2_score(y_test, y_pred_poly),
        r2_score(y_test, y_pred_ada),
        r2_score(y_test, y_pred_tree),
        r2_score(y_test, y_pred_et),
        r2_score(y_test, y_pred_svr),
        r2_score(y_test, y_pred_gb)
    ],
    'MAE': [
        mean_absolute_error(y_test, y_pred_linear),
        mean_absolute_error(y_test, y_pred_poly),
        mean_absolute_error(y_test, y_pred_ada),
        mean_absolute_error(y_test, y_pred_tree),
        mean_absolute_error(y_test, y_pred_et),
        mean_absolute_error(y_test, y_pred_svr),
        mean_absolute_error(y_test, y_pred_gb)
    ],
    "Accuracy in %":[
        (abs(y_test - y_pred_linear) <= tolerance).mean()*100,
        (abs(y_test - y_pred_poly) <= tolerance).mean()*100,
        (abs(y_test - y_pred_ada) <= tolerance).mean()*100,
        (abs(y_test - y_pred_tree) <= tolerance).mean()*100,
        (abs(y_test - y_pred_et) <= tolerance).mean()*100,
        (abs(y_test - y_pred_svr) <= tolerance).mean()*100,
        (abs(y_test - y_pred_gb) <= tolerance).mean()*100,
    ]
})

print(results)
```

```
                          Model  R2 Score         MAE  Accuracy in %
0                        Linear  0.338272  175.998586           12.5
1          Polynomial Regression  0.248061  181.244829           25.0
2                      AdaBoost  0.940735   37.125000           62.5
3                 Decision Tree  0.837528   64.009615           50.0
4                   Extra Trees  0.950903   44.308093           37.5
5     Support-Vector Regression -0.476629  208.337355           50.0
6  Gradient-Boosting Regression  0.980193   18.030025           87.5
```

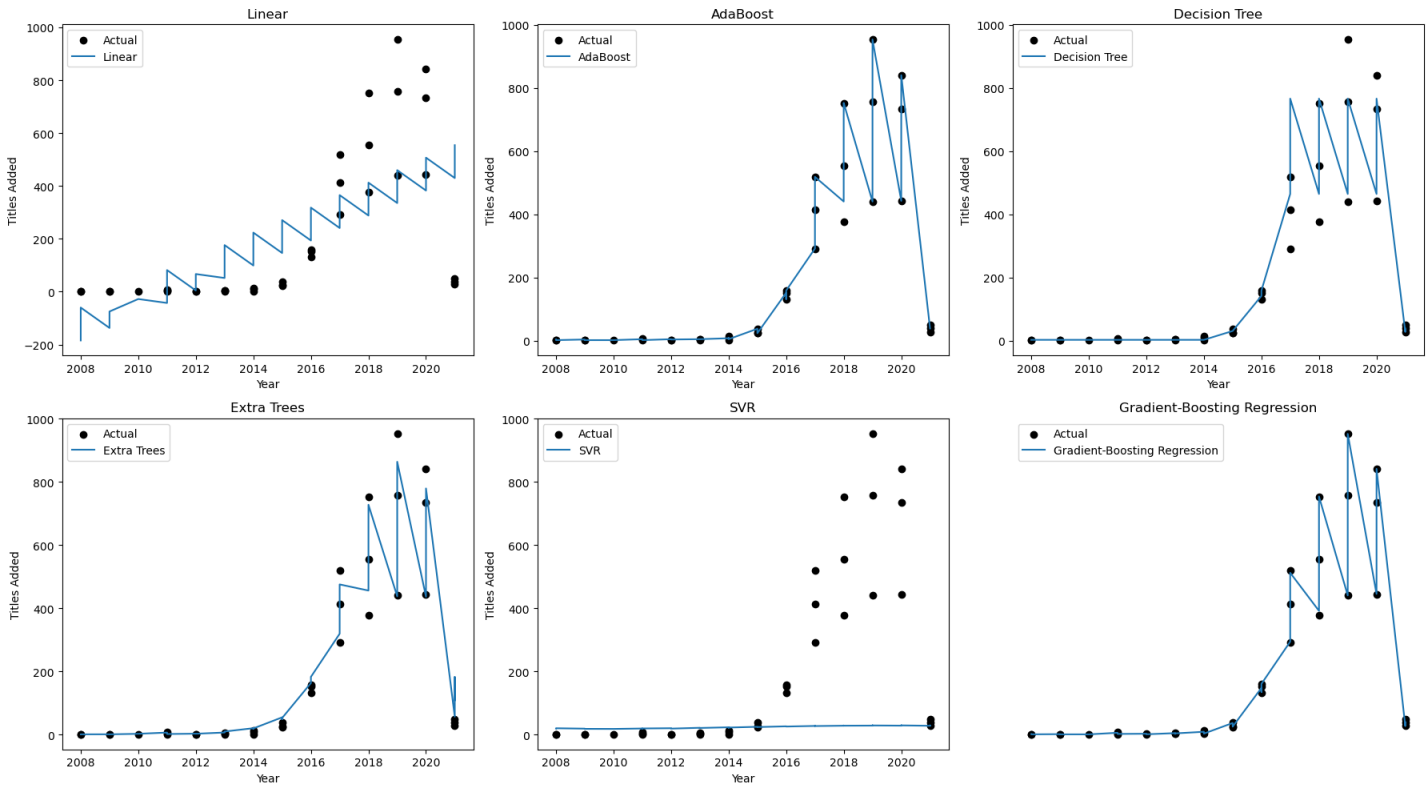
CONCLUSION]

Tree-based models such as DecisionTreeRegressor, ExtraTreesRegressor, AdaBoostRegressor, and GradientBoostingRegressor achieved high accuracy on test data, with Gradient Boosting giving the best performance in terms of R^2 score and MAE.

In [ ]:

In [53]:

```
# Sort for smooth plotting
sorted_idx = X['year_added'].argsort()
X_sorted = X.iloc[sorted_idx]

fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()  # flatten to access easily

models = [
    ('Linear', linear_pipeline),
    ('AdaBoost', ada_pipeline),
    ('Decision Tree', tree_pipeline),
    ('Extra Trees', et_pipeline),
    ('SVR', svr_pipeline),
    ('Gradient-Boosting Regression',gb_pipeline)
]

for i, (name, model) in enumerate(models):
    ax = axes[i]

    # Scatter (actual)
    ax.scatter(X['year_added'], y, color='black', label='Actual')

    # Line (prediction)
    ax.plot(X_sorted['year_added'], model.predict(X_sorted), label=name)

    ax.set_title(name)
    ax.set_xlabel('Year')
    ax.set_ylabel('Titles Added')
    ax.legend()

# Hide the 6th empty subplot
axes[5].axis('off')

plt.tight_layout()
plt.show()
```



In [ ]:

Conclusion: Forecasting Netflix Titles (2022–2025)

After comparing regression models, we see a wide divergence in forecasts:

- **Linear Regression** projects steady growth, from ~536 titles in 2022 to ~677 titles in 2025.
- **AdaBoost, Decision Tree, Extra Trees, SVR, and Gradient Boosting** remain nearly flat, predicting between 25–39 titles per year, showing limited sensitivity to historical trends.

Projected Titles Added (2022–2025)

Based on the table of model outputs:

Year	Linear	Polynomial	AdaBoost	Decision Tree	Extra Trees	SVR	Gradient Boosting
2022	536	639	39	28	177	28	39
2023	583	721	39	28	177	27	39
2024	630	808	39	28	177	26	39
2025	677	899	39	28	177	25	39

Key Takeaway

- **Linear Regression** suggests Netflix will continue adding **~140 more titles over 4 years** and **Polynomial Regression** suggests **~260 more titles over 4 years**.
- Other models (AdaBoost, SVR, Gradient Boosting) predict **minimal growth**, essentially flatlining around 25–40 titles annually.
- **Extra Trees** sits in between, holding steady at ~177 titles per year.

Overall, the **Linear Regression model provides the most realistic growth trajectory** , while the others appear underfit or overly constrained. If we trust the linear trend, Netflix's catalog expansion will be moderate but consistent through 2025.

In [ ]:

In [ ]:

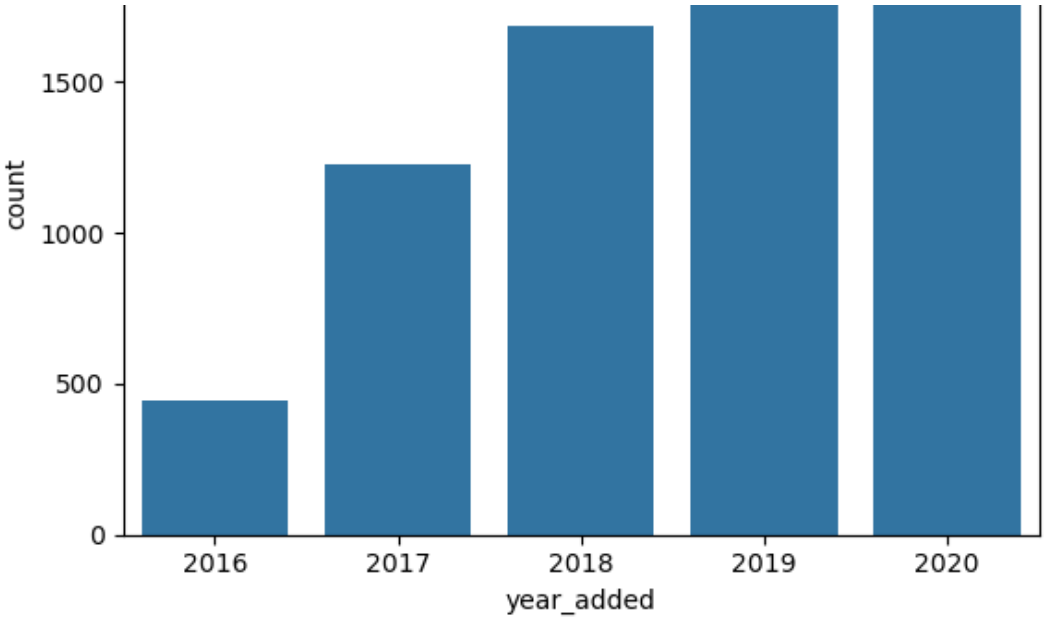
In [ ]:

In [ ]:

Q7] How is Netflix content growing over time?

In [54]:

```
data = df["year_added"].value_counts().reset_index()
data = data.head(5)
sns.barplot(x=data["year_added"],y=data["count"])

plt.show()
```



In [ ]:

```

```

In [55]:

```
yearly_data.sort_values(by="title_count",ascending=False).head(5)
```

Out[55]:

	year_added	genre_count	title_count
30	2019	3	954
33	2020	3	841
29	2019	2	758
27	2018	3	753
32	2020	2	735

CONCLUSION] Netflix content has shown a consistent upward trend over the years, with significant growth in recent years. Linear models were able to capture this trend effectively, while tree-based models failed due to their inability to extrapolate beyond training data. This highlights the importance of selecting models based on problem type rather than complexity.

In [ ]:

```

```

In [ ]:

```

```

Q8] Who are the top contributing directors?

In [56]:

```
data = df["director"].value_counts().reset_index().head()
data = data[data["director"]!="Unknown"]
data
```

Out[56]:

	director	count

	director	count
1	Raúl Campos, Jan Suter	18
2	Marcus Raboy	16
3	Jay Karas	14
4	Cathy Garcia-Molina	13

CONCLUSION] The top contributing directors include:

1) Raúl Campos 2) Jan Suter 3) Marcus Raboy 4) Jay Karas 5) Cathy Garcia-Molina

In [ ]:

```

```

Q9] Which countries contribute most to the content-library?

In [57]:

```
data = df["first_country"].value_counts().reset_index().head()
data = data[data["first_country"]!="Unknown"]
data
```

Out[57]:

	first_country	count
0	United States	2883
1	India	956
2	United Kingdom	577
4	Canada	259

CONCLUSION] The top contributing countries are:

1) United States 2) India 3) United Kingdom 4) Canada

In [ ]:

```

```

In [ ]:

```

```

Q10] Is there a statistically significant difference in the average duration of Netflix content between Movies and TV Shows?

In [58]:

```
from scipy.stats import ttest_ind


# Split data into two groups
movies = df[df['type'] == 'Movie']['duration_value']
tv_shows = df[df['type'] == 'TV Show']['duration_value']

# Perform Independent T-Test
t_stat, p_value = ttest_ind(movies, tv_shows, equal_var=False)

# Print results
print("T-Statistic:", t_stat)
print("P-Value:", p_value)
```

```
# Decision
alpha = 0.05

if p_value < alpha:
    print("\nReject Null Hypothesis (H0)")
    print("There is a statistically significant difference in duration.")
else:
    print("\nFail to Reject Null Hypothesis (H0)")
    print("No significant difference in duration.")
```

```
T-Statistic: 249.79900942857398
P-Value: 0.0

Reject Null Hypothesis (H0)
There is a statistically significant difference in duration.
```

CONCLUSION] There is a significant difference in the durations of tv-shows and movies.

In [ ]:

```

```

Q11] Is the proportion of TV Shows significantly higher in recent years (after 2018) compared to older years?

In [59]:

```
import pandas as pd
import numpy as np
from statsmodels.stats.proportion import proportions_ztest

# Create groups
recent = df[df['release_year'] >= 2018]
old = df[df['release_year'] < 2018]

# Count TV shows
success_recent = (recent['type'] == 'TV Show').sum()
success_old = (old['type'] == 'TV Show').sum()

# Total samples
n_recent = len(recent)
n_old = len(old)

# Z-test
count = np.array([success_recent, success_old])
nobs = np.array([n_recent, n_old])

z_stat, p_value = proportions_ztest(count, nobs, alternative='larger')

print("Z-stat:", z_stat)
print("P-value:", p_value)

# Decision
alpha = 0.05
if p_value < alpha:
    print("Reject H0: TV Shows proportion increased in recent years")
else:
    print("Fail to Reject H0")
```

```
Z-stat: 17.28945184862103
P-value: 2.8243427889897262e-67
Reject H0: TV Shows proportion increased in recent years
```

In [ ]:

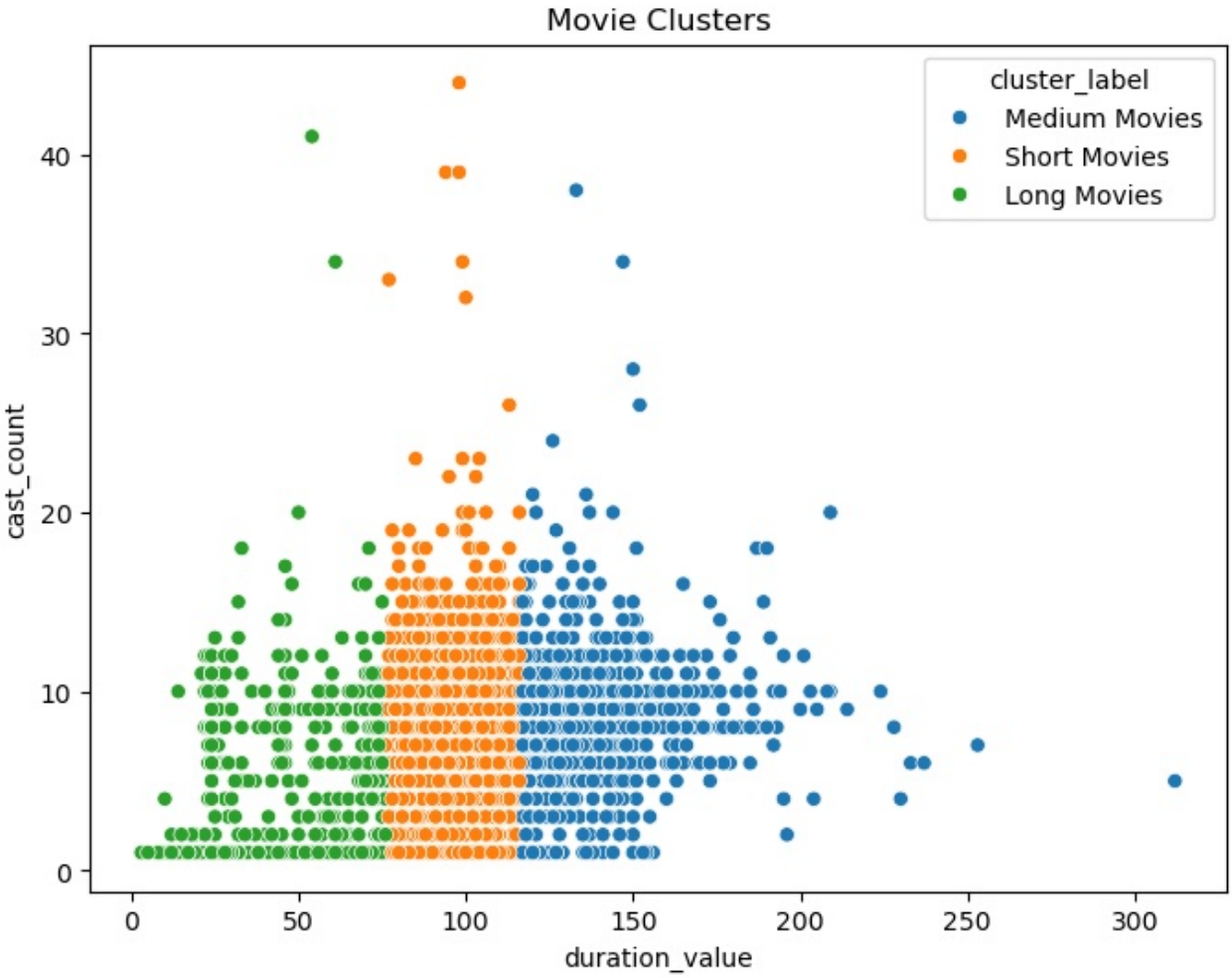
```

```

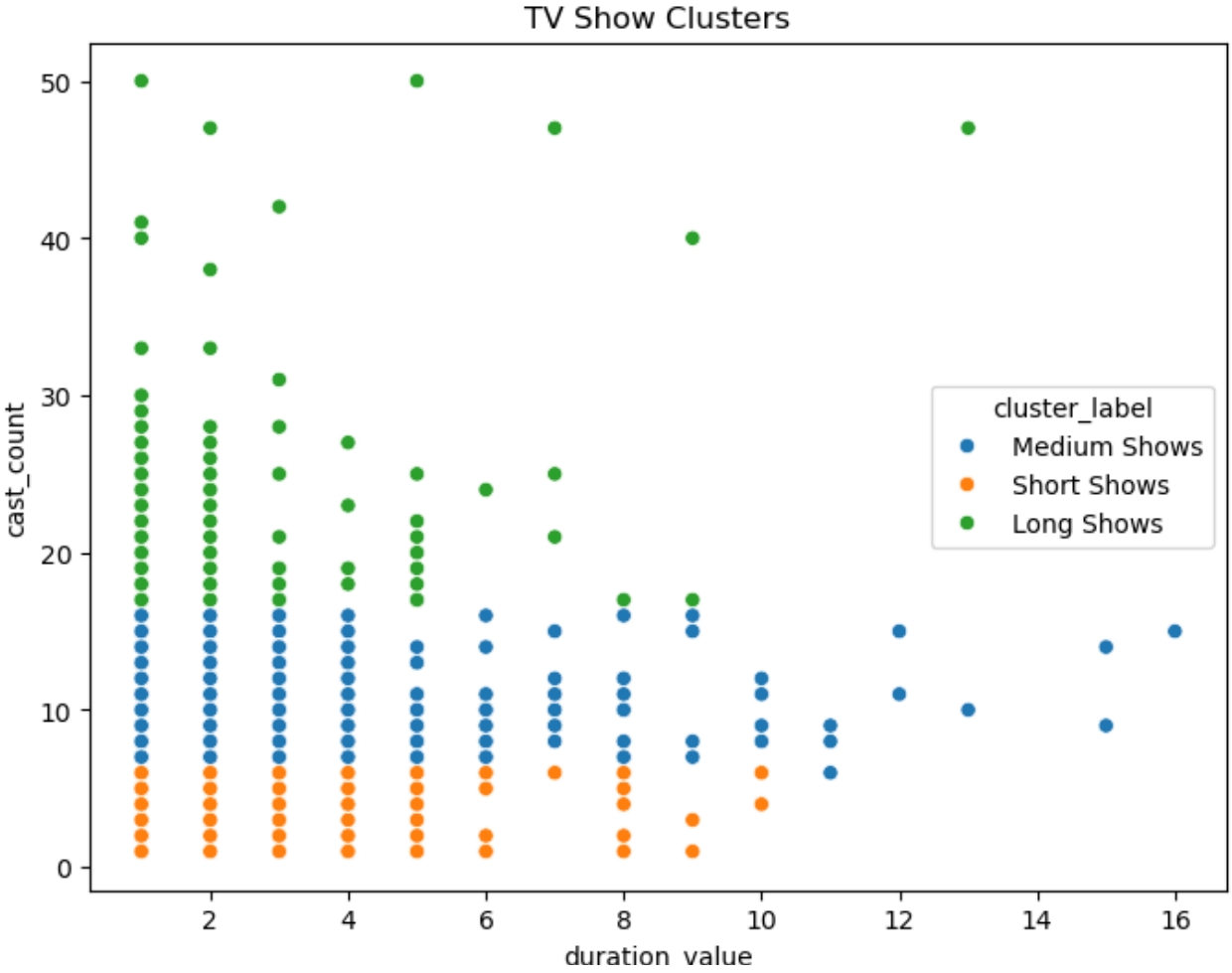
Q12] Can Netflix content be grouped into meaningful clusters based on features like genre, cast size, and duration?

In [60]:

```
from sklearn.cluster import KMeans



movies_df = df[df['type'] == 'Movie'].copy()
tv_df = df[df['type'] == 'TV Show'].copy()


features = ['duration_value', 'cast_count', 'genre_count']









kmeans_movies = KMeans(n_clusters=3, random_state=42)
movies_df['cluster'] = kmeans_movies.fit_predict(movies_df[features])

movie_labels = {
    0: "Short Movies",
    1: "Medium Movies",
    2: "Long Movies"
}
movies_df['cluster_label'] = movies_df['cluster'].map(movie_labels)









kmeans_tv = KMeans(n_clusters=3, random_state=42)
tv_df['cluster'] = kmeans_tv.fit_predict(tv_df[features])

tv_labels = {
    0: "Short Shows",
    1: "Medium Shows",
    2: "Long Shows"
}
tv_df['cluster_label'] = tv_df['cluster'].map(tv_labels)









# Movies Plot
plt.figure(figsize=(8,6))
sns.scatterplot(
    x='duration_value',
    y='cast_count',
    hue='cluster_label',
    data=movies_df
)
plt.title("Movie Clusters")
plt.show()

# TV Shows Plot
plt.figure(figsize=(8,6))
sns.scatterplot(
    x='duration_value',
```

```
    y='cast_count',
    hue='cluster_label',
    data=tv_df
)
plt.title("TV Show Clusters")
plt.show()
```





In [61]:

```
print("\n--- MOVIE CLUSTERS ---")
for label in movies_df['cluster_label'].unique():
    subset = movies_df[movies_df['cluster_label'] == label]
    print(label)
    print("Avg Duration:", subset['duration_value'].mean())
    print("Avg Cast:", subset['cast_count'].mean())

print("\n--- TV SHOW CLUSTERS ---")
for label in tv_df['cluster_label'].unique():
    subset = tv_df[tv_df['cluster_label'] == label]
    print(label)
    print("Avg Duration:", subset['duration_value'].mean())
    print("Avg Cast:", subset['cast_count'].mean())
```

```
--- MOVIE CLUSTERS ---
Medium Movies
Avg Duration: 136.6070549630845
Avg Cast: 8.71862182116489
Short Movies
Avg Duration: 96.86318332829961
Avg Cast: 7.569918453639384
Long Movies
Avg Duration: 55.184179456906726
Avg Cast: 3.5962219598583234

--- TV SHOW CLUSTERS ---
Medium Shows
Avg Duration: 1.9271012006861064
Avg Cast: 10.01114922813036
Short Shows
Avg Duration: 1.5527502254283139
Avg Cast: 2.8872858431018935
Long Shows
Avg Duration: 2.303703703703704
Avg Cast: 22.666666666666668
```

Q13] Which genres are overcrowded vs underserved?

In [62]:

```
from collections import Counter

new_df = df.copy()
new_df['genre_list'] = new_df['genres'].apply(lambda x: x.split(','))

all_genres = [g.strip() for sublist in new_df['genre_list'] for g in sublist]

genre_counts = Counter(all_genres)

genre_df = pd.DataFrame(genre_counts.items(), columns=['genre', 'count'])
genre_df = genre_df.sort_values(by='count', ascending=False)
```

In [63]:

```
print("Overcrowded genres:")
genre_df.head(10)
```

```
Overcrowded genres:
```

Out[63]:

	genre	count
4	International Movies	2437
6	Dramas	2106

	genre	count
8	Comedies	1471
0	International TV Shows	1199
15	Documentaries	786
13	Action & Adventure	721
1	TV Dramas	704
12	Independent Movies	673
21	Children & Family Movies	532
14	Romantic Movies	531

In [64]:

```
print("Underserved genres:")
genre_df.tail(10)
```

```
Underserved genres:
```

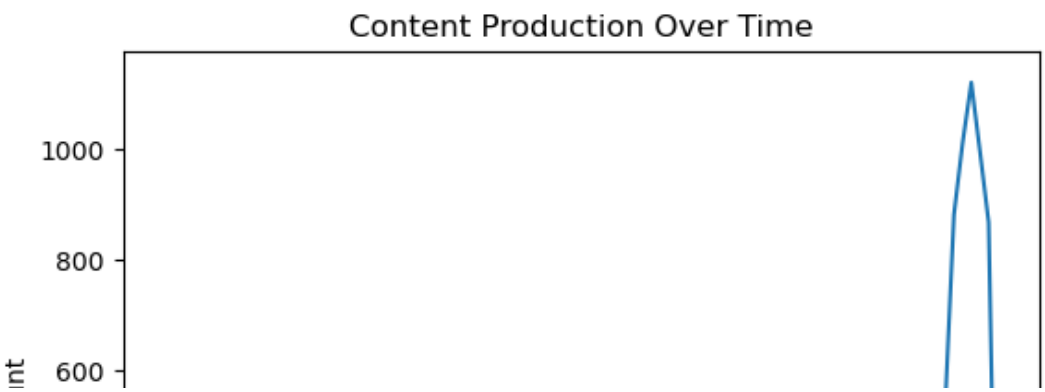
Out[64]:

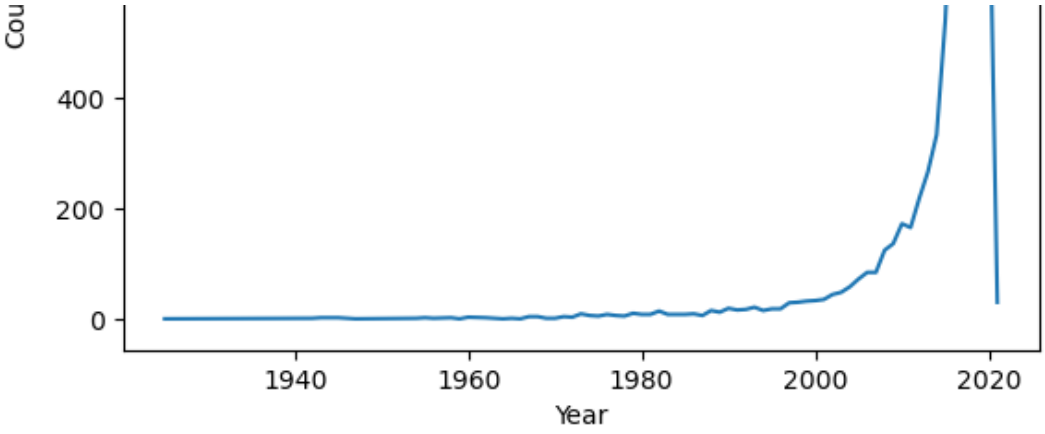
	genre	count
38	TV Horror	69
36	Teen TV Shows	60
37	Cult Movies	59
23	Faith & Spirituality	57
41	Anime Features	57
26	Movies	56
35	Stand-Up Comedy & Talk Shows	52
33	TV Thrillers	50
40	Classic & Cult TV	27
20	TV Shows	12

Q14] How has content production evolved over time?

In [65]:

```
import matplotlib.pyplot as plt

df['release_year'] = pd.to_numeric(df['release_year'], errors='coerce')

yearly = df.groupby('release_year').size()

yearly.plot(kind='line')
plt.title("Content Production Over Time")
plt.xlabel("Year")
plt.ylabel("Count")
plt.show()
```





CONCLUSION] With this we can say that most content-addition was done between the years 2000 and 2020.

In [ ]:

```

```

In [ ]:

```

```

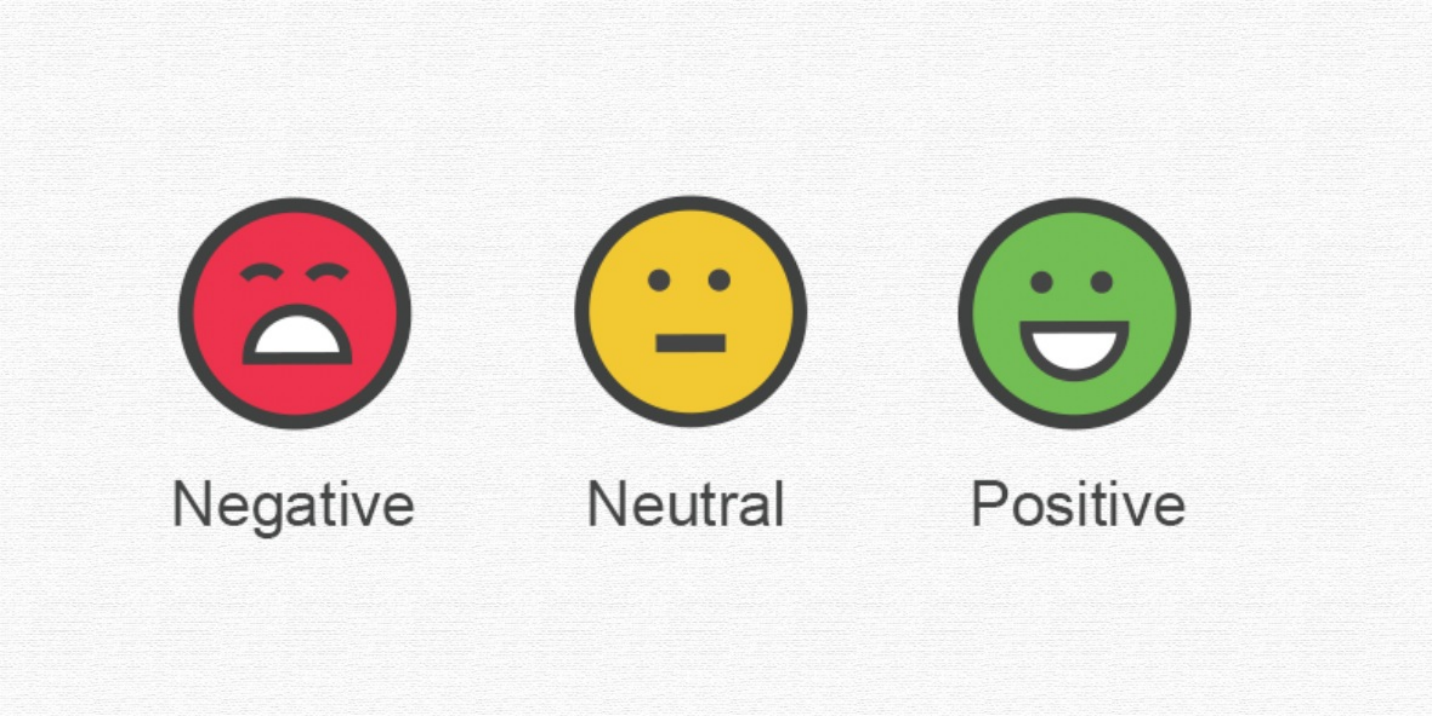
In [ ]:

```

```

Q15] Now label sentiment for every content:

In [66]:

```
from IPython.display import Image, display
display(Image(filename="sentiments.png"))
```



In [67]:

```
import nltk
from nltk.sentiment import SentimentIntensityAnalyzer
from nltk.corpus import stopwords
from nltk.tokenize import word_tokenize
```

```
from nltk.stem import WordNetLemmatizer

nltk.download('vader_lexicon')
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
nltk.download('omw-1.4')

stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()
analyzer = SentimentIntensityAnalyzer()

def preprocess_text(text):
    text = str(text).lower()
    tokens = word_tokenize(text)
    filtered_tokens = [word for word in tokens if word.isalnum() and word not in stop_wo
rds]
    lemmatized_tokens = [lemmatizer.lemmatize(word) for word in filtered_tokens]
    return " ".join(lemmatized_tokens)

def get_sentiment(text):
    scores = analyzer.polarity_scores(text)
    compound = scores['compound']
    if compound >= 0.05:
        return 'Positive'
    elif compound <= -0.05:
        return 'Negative'
    else:
        return 'Neutral'

df['content_sentiment_text'] = df['description'].fillna('').apply(preprocess_text)
df['sentiment'] = df['content_sentiment_text'].apply(get_sentiment)

df[['title', 'description', 'content_sentiment_text', 'sentiment']].head()
```

```
[nltk_data] Downloading package vader_lexicon to
[nltk_data]     C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data]   Package vader_lexicon is already up-to-date!
[nltk_data] Downloading package punkt to
[nltk_data]     C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data]   Package punkt is already up-to-date!
[nltk_data] Downloading package stopwords to
[nltk_data]     C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data]   Package stopwords is already up-to-date!
[nltk_data] Downloading package wordnet to
[nltk_data]     C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data]   Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data]     C:\Users\Lenovo\AppData\Roaming\nltk_data...
[nltk_data]   Package omw-1.4 is already up-to-date!
```

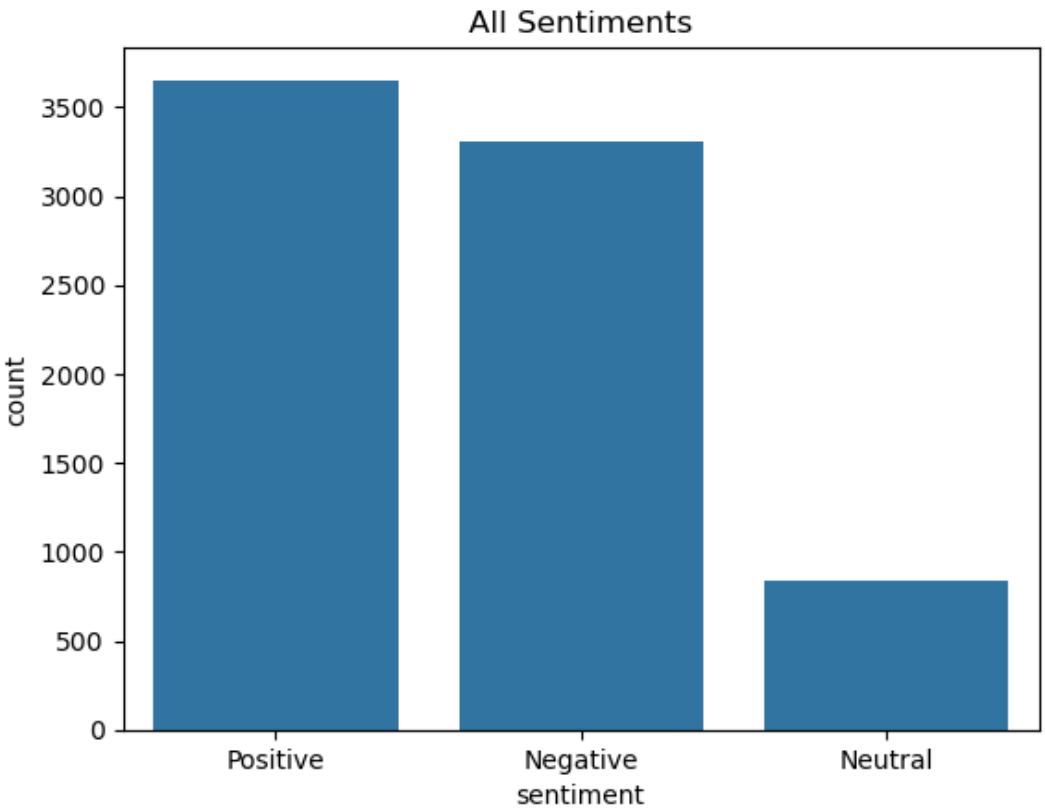
Out[67]:

	title	description	content_sentiment_text	sentiment
0	3%	In a future where the elite inhabit an island ...	future elite inhabit island paradise far crowd...	Positive
1	1920	An architect and his wife move into a castle t...	architect wife move castle slated become luxur...	Positive
2	3 Heroines	Three Indonesian women break records by becomi...	three indonesian woman break record becoming f...	Positive
3	Blue Mountain State: The Rise of Thadland	New NFL star Thad buys his old teammates' belo...	new nfl star thad buy old teammate beloved fra...	Positive
4	Blue Planet II	This sequel to the award-winning nature series...	sequel nature series blue planet dive beneath ...	Positive

In [ ]:

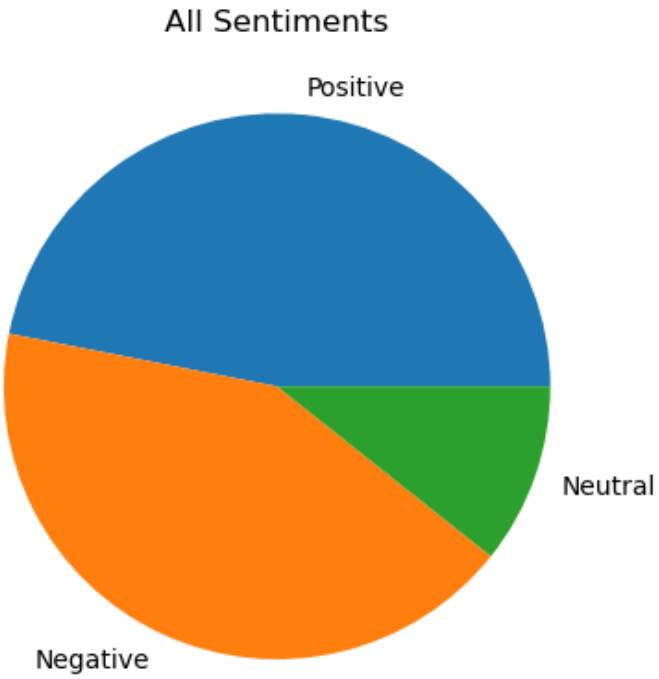
In [68]:

```
import seaborn as sns
import matplotlib.pyplot as plt


plt.title("All Sentiments")
sns.countplot(x='sentiment', data=df, order=df['sentiment'].value_counts().index)

plt.show()
```



In [69]:

```
import matplotlib.pyplot as plt
plt.pie(df["sentiment"].value_counts(),labels=df["sentiment"].value_counts().index)
plt.title("All Sentiments")
plt.show()
```



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